

CE-320 REINFORCED CONCRETE DESIGN -I



COMPLEX ENGINEERING PROBLEM REPORT

DEPARTMENT OF CIVIL ENGINEERING

**NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY KARACHI,
PAKISTAN**

RCD REPORT

BATCH: 2022

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PAKISTAN**

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REFERENCES

LIST OF CODES

- ACI Code, Section 25.7.3.2
- ACI Code, Section 25.7.3.1
- ACI Code, Section 25.7.3.6
- ACI Code, Section 25.7.3.6
- ACI Code, Section 19.2.4.1
- ACI Code, Section 19.2.4.2
- ACI Code, Section 19.2.4.3
- ACI Code, Section 24.2.3.5
- ACI Code Section 24.2.2
- ACI Code, Section 9.6.3.4
- ACI Code, Section 22.5.5.1

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LIST OF ABBREVIATIONS

In	Inches
K	Kips
Psi	per square foot
Ksi	kilo pound per square inch
Fy	Specified yield strength of steel reinforcement
Mu	Moment strength due to factored loads
qu	Ultimate bearing capacity of soil using factor load
Vc	Shear strength of concrete
Vu	Shear force due to factored loads

CHAPTER NO. 1

INTRODUCTION

1.1 PROBLEM STATEMENT

You are working as a junior structural engineer in a design firm, and you have been assigned to complete the preliminary design of the N BLOCK located at the main campus of NED UET. The contractor is asking you to provide the following set of drawings for the execution of construction work.

1.2 REQUIRED

a) Architectural Drawing

- a. Plan
- b. Section
- c. Elevation

b) Structural Detailing Drawing

- a. Beam Framing Plan
- b. Beam Detailing Drawing
- c. Column Detailing Drawing
- d. Slab Detailing Drawing
- e. Footing Detailing Drawing

1.3 OBJECTIVE OF THE PROJECT

Following are the objectives of the project:

- You can refer to figure 1 as reference for the building to be designed for this assignment.
- Use AUTOCAD for the preparation of architectural and structural drawings.
- For structural analysis use any software like GRASP, ETAB, or SAP.
- Present all calculations in the final report.
- Use A3 pages for all drawings.

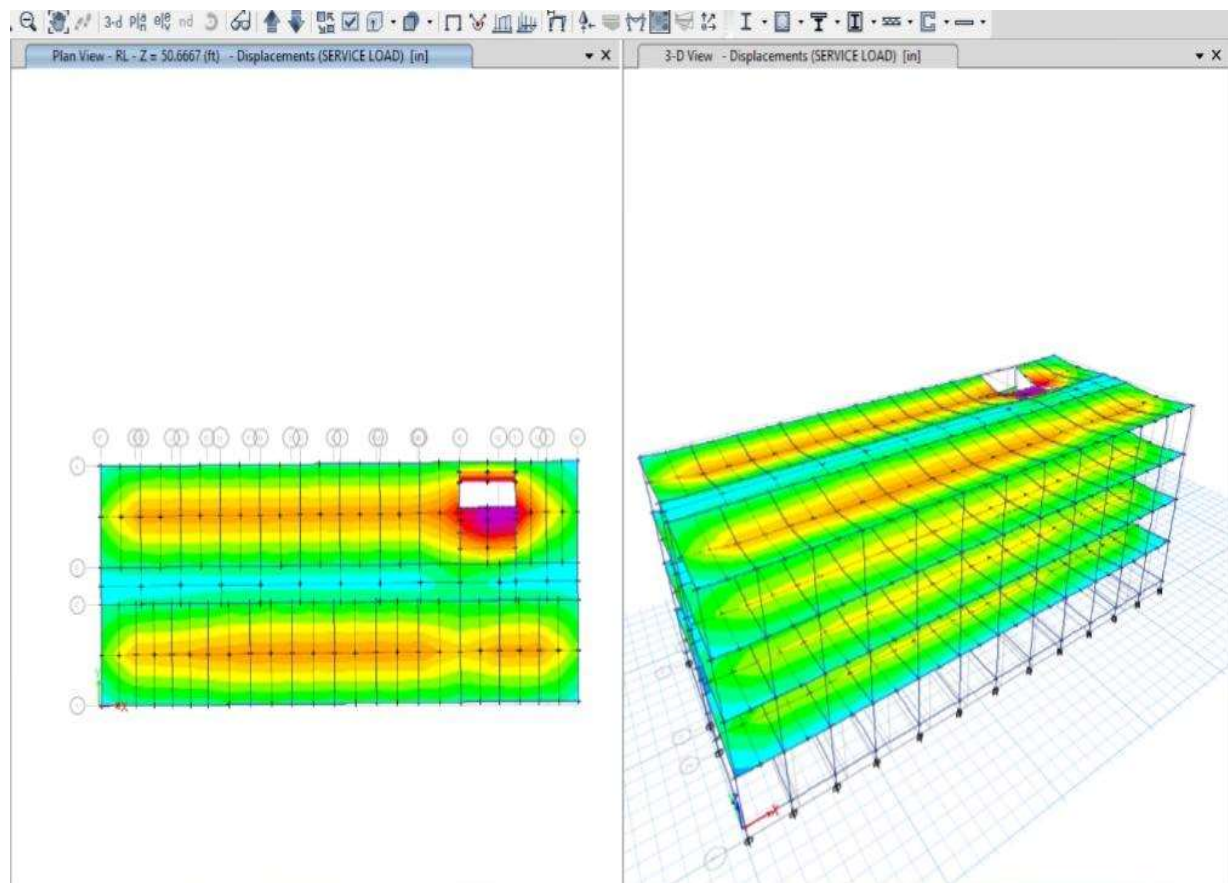
1.4 SCOPE

The scope of this complex engineering problem is to design a column, slab, beam and footing

1.5 EXPECTED OUTCOMES

The expected outcomes are as follow

- Architectural drawing prepared according to the requirement.
- Advanced structural analysis detailing with complete details by using software.
- Detailed design calculations as per requirement.
- Structural detailing is prepared.



CHAPTER NO. 2

LITERATURE REVIEW

2.1 GENERAL

a) Reinforced Concrete:

Reinforced Concrete is the common term given to a concrete member (or slab) that contains steel reinforcement (usually in the form of steel bars) to increase the strength of the structure. The material that results from the combination of concrete and reinforcing bars is called Reinforced Concrete (RC). During construction, the reinforcement steel is placed in the formwork first, either in the form of a prefabricated steel cage or steel rebar that are fastened together and wired in-situ. Then concrete is poured into the formwork and vibrated using adequate devices so as to guarantee a high level of collaboration between the two materials.

b) Importance of Reinforced Concrete:

Reinforced concrete has very good strength in tension as well as compression. This makes concrete a desired construction material. Concrete constituents are widely available worldwide and inexpensive. Similarly, the production cost of concrete is very low. There is an overall economy by using reinforced concrete because its maintenance cost is low due to the long-lasting nature of reinforced concrete. Due to low permeability, concrete can resist chemicals dissolved in water such as sulfates, chloride and carbon dioxide, which may cause corrosion in concrete, without serious deterioration. Reinforced concrete buildings are more fire resistant than other commonly used construction materials like steel and wood. Ductility enables concrete to show sign of distress such as cracking and deflection if reinforced concrete member experience overloading. This enables engineers to consider suitable measures to prevent further concrete damages.

Properly designed reinforced concrete structures are extremely resistant to earthquakes.

c) Applications of Reinforced Concrete

- a) Buildings
- b) Roads
- c) Foundations
- d) Precast Works
- e) Retaining Walls
- f) Bunkers and Silos

2.2 Literature Collection:

This chapter comprises of a discussion about the relevant literature collected for the reinforced design of the column, beams, slabs and footing of the new building that has been constructed in the civil department in order to design them we require few like the dimension of the building and architectural drawing along with its structural drawing and remember the code of ACI (AMERICAN CONCRETE INSTITUTE). It is the code which gives us the limit of shear and design and it is followed all around the world.

a) COLUMN:

A column or pillar in architecture and structural engineering is a structural element that transmits, through compression, the weight of the structure above to other structural elements below.

ACI specifications

- The Steel percentage range is from 1% to 8% of the gross area of the section (ACI Code, Section 10.6.1.1)
- Minimum reinforcement is necessary to provide resistance to bending, which may exist, and to reduce the effects of creep and shrinkage of the concrete under sustained compressive stresses.
- Practically, it is very difficult to fit more than 8% of steel reinforcement into a column and maintain sufficient space for concrete to flow between bars.
- In cases where the area of the column provided larger than the area required it shall be permitted to have a minimum area of steel of 0.5% of the gross cross sectional area.
- At least four bars are required for tied circular and rectangular members
- Six bars are needed for circular members enclosed by spirals (ACI Code, Section 10.7.3.1).
- For other shapes, one bar should be provided at each corner, and proper lateral reinforcement must be provided.
- For tied triangular columns, at least three bars are required.
- Bars shall not be located at a distance greater than 6 in. clear on either side from a laterally supported bar.

b) BEAMS

Beams are usually horizontal structural elements that carry loads perpendicular to their longitudinal direction

c) SHEAR

Shear is opposing structural forces, which causes slippage, on a plane and failure.

ACI specifications

- Flexure-Shear Cracks the ACI Code specifies a capacity reduction factor, $\phi = 0.75$ for shear.
- In general, the factored shear strength varies between 0 fc' and 5 fc' as per ACI Code
- A minimum area of shear reinforcement is specified by the Code. The ACI Code, Section 9.6.3.4,
- The presence of large moments on a beam, for which adequate reinforcement is provided, the nominal shear force at which diagonal tension cracks develop is given by ACI Code, Section 22.5.5.1.

d) FOR FOOTING

The footing is a structural member that safely transmits the load of the upper building to the ground.

e) FOR SLABS

A concrete slab is a structural feature, usually of constant thickness, that can be used as a floor or a roof. A slab-on-ground is supported on the subsoil and is usually reinforced with reinforcing bars

f) SHEAR DIAGRAM

A shear force diagram is one which shows variation in shear force along the length of the beam.

g) BENDING DIAGRAM

Bending moment may be defined as "the sum of moments about that section of all external forces acting to one side of that section

h) SOFTWARES USED



Architectural Drawing:

- GROUND FLOOR PLAN:**

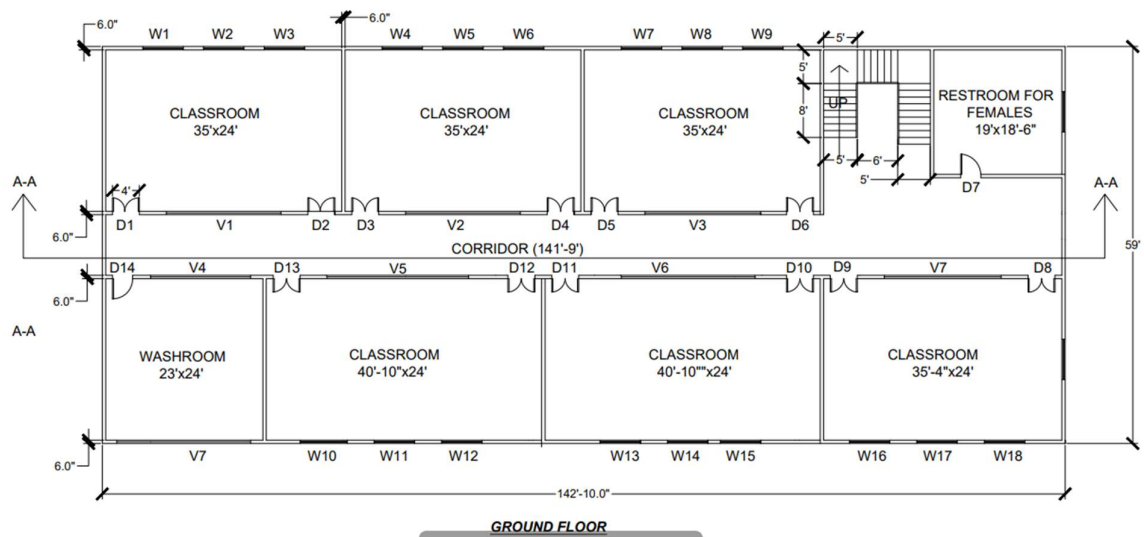


FIGURE 1

- FIRST FLOOR PLAN:**

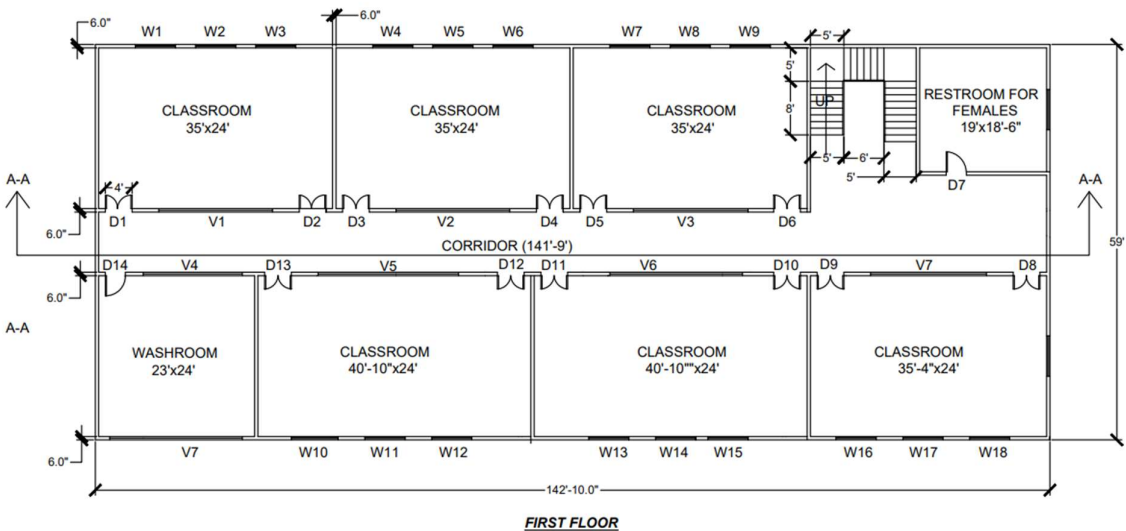


FIGURE 2

- **ELEVATION:**

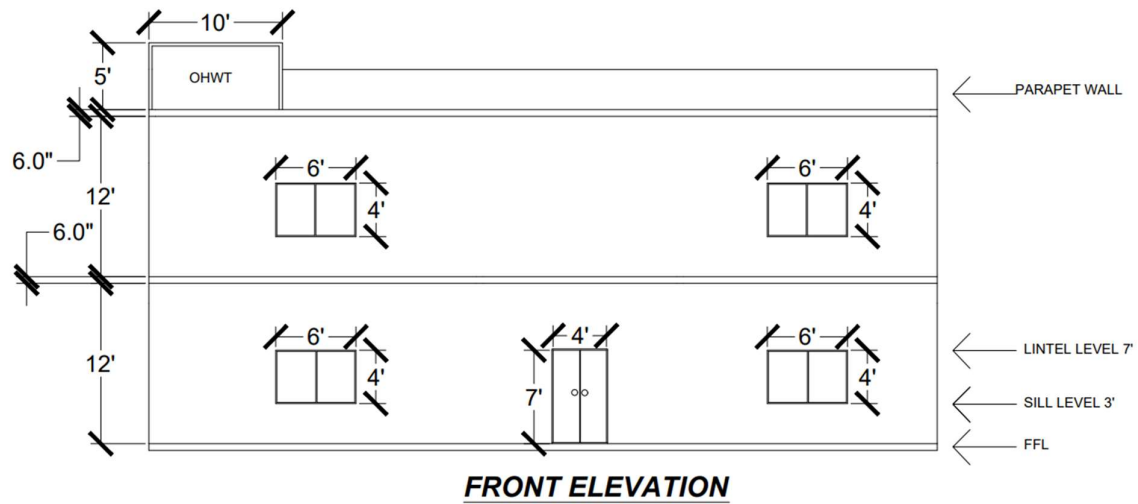


FIGURE 3

- **SECTION:**

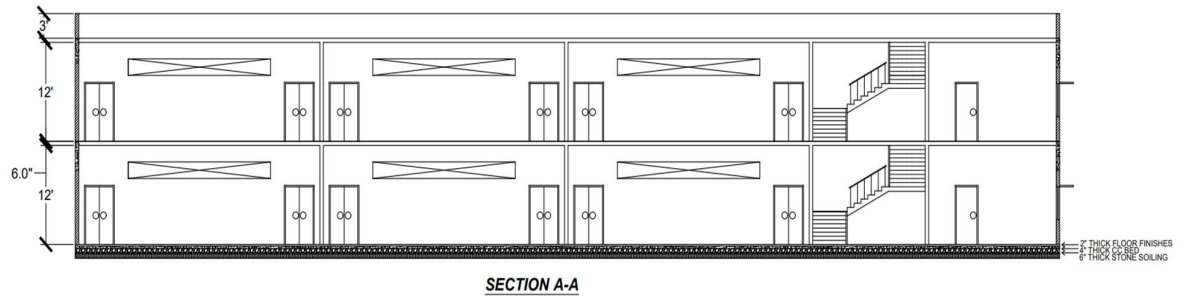


FIGURE 4

CHAPTER NO. 3

METHODOLOGY:

Design of Column:

- **Method:**

- Evaluate the factored axial load $u P$ acting on the column. Decide on a reinforcement ratio ρ that satisfies ACI Code limits.
- Usually a 1 % ratio is chosen for economic considerations. Determine the gross sectional area A_g of the concrete section.
- Choose the dimensions of the cross section based on its shape. For rectangular sections, the ratio of the longer to shorter side is recommended to not exceed 3.
- Readjust the reinforcement ratio by substituting the actual cross sectional area.
- This ratio has to fall within the specified code limits.
- Calculate the needed area of longitudinal reinforcement ratio based on the adjusted reinforced ratio and the chosen concrete dimensions.
- From reinforcement tables, choose the number and diameters of needed reinforcing bars. For rectangular sections, a minimum of four bars is needed, while a minimum of six bars is used for circular columns.
- Design the lateral reinforcement according to the type of column, either ties or spirals,
- Check whether the spacing between longitudinal reinforcing bars satisfies ACI Code requirements.
- Draw the designed section showing concrete dimensions and with required longitudinal and lateral reinforcement.
- We have design three column by following this step and then have calculate largest moment where the load is larger by using Etabs

- **Formula:**

- $P_u = \phi P_n = \phi k [0.85f_c' A_g - A_{st} + A_{st}f_y]$
- $A_{bar} = \pi r^2$
- $\rho = \frac{A_{st}}{A_g}$
- Ultimate Load = $1.2D + 1.6L$
- SPACING OF TIES
- 16 times longitudinal bar diameter.
- 48 times tie diameter.
- Least cross sectional dimension 6 inches

Load Calculation (Catchment Area Method) (CORNER & EDGE COLUMN)

Catchment Area = 5.945' x 12.335'

Self-Weight of Slab:

$$\begin{aligned} &= (150 \text{ lb/ft}^3) \times (8/12) \\ &= 100 \end{aligned}$$

Finishes Load:

$$\begin{aligned} &= 144 \text{ lb/ft}^3 \times (3/12) \\ &= 36 \text{ lb/ft}^2 \end{aligned}$$

Masonry Load:

$$\begin{aligned} &= (144 \text{ lb/ft}^3) \times (12.67 - 2.67) \times (0.5) \times (12.335) \\ &= 8881.2 \text{ lb} \end{aligned}$$

Self-Weight of Beam

$$\begin{aligned} &= 150 \text{ lb/ft}^3 \times (8/12) \times (32/12) \times 12.335 \\ &= 3289.33 \text{ lb} \end{aligned}$$

Self-Weight of Column

$$\begin{aligned} &= 150 \text{ lb/ft}^3 \times (12) \times (12) \times (50.67) \\ &= 7600.5 \text{ lb} \end{aligned}$$

Total Dead Load:

$$\begin{aligned} &= 100 + 36 \\ &= 136 \text{ lb/ft}^2 \end{aligned}$$

Point Load:

$$\begin{aligned} &= 136 \text{ lb/ft}^2 \times 5.945 \text{ ft} \times 12.335 \text{ ft} \\ &= 9973.09 \text{ lb} \end{aligned}$$

Final Dead Load (adding masonry load and self wt of beam and column)

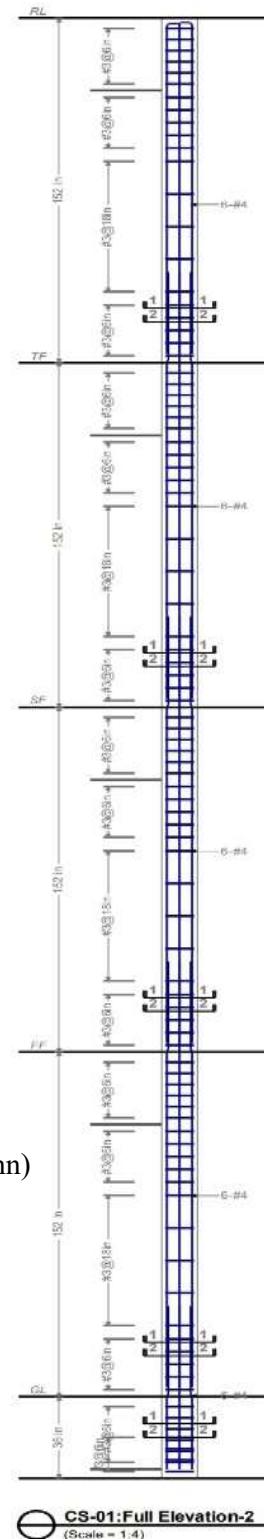
$$\begin{aligned} &= 9973.09 + 3289.33 + 8881.2 + 7600.5 \\ &= 29,744.12 \text{ lb or } 29.74 \text{ kips} \end{aligned}$$

Live Load

Live Load (classroom) = 60 lb/ft²

To Point Load:

$$\begin{aligned} &= 60 \text{ lb/ft}^2 \times 5.945' \times 12.335' \\ &= 4,399.89 \text{ lb or } 4.399 \text{ kips} \end{aligned}$$



Ultimate Load

$$P_u = 1.2 W_o + 1.6 W_c$$

$$= (1.2 \times 29.74) + (1.6 \times 4.399)$$

$$= 42.72 \text{ kip}$$

Considering $n = 4$, G+3

$$P_u = 42.72 \times 4$$

$$P_u = 170.88 \text{ kip}$$

Procedure:

We have use the catchment method to determine the load

Given:

- $r_{\text{concrete}} = 150 \text{ pcf}$
- $r_{\text{reinforced}} = 144 \text{ pcf}$
- Slab thickness = 8"
- Live Load for classrooms = 60 psf
- Ultimate load = $1.2P_d + 1.6P_L$
- Ultimate load = 170.88 Kip

Assume:**For tied columns:**

$$\Phi = 0.65$$

$$k=0.8$$

$$f_y = 60 \text{ ksi or } 60000 \text{ psi}$$

$$f'_c = 4 \text{ ksi or } 4000 \text{ psi}$$

Equation:

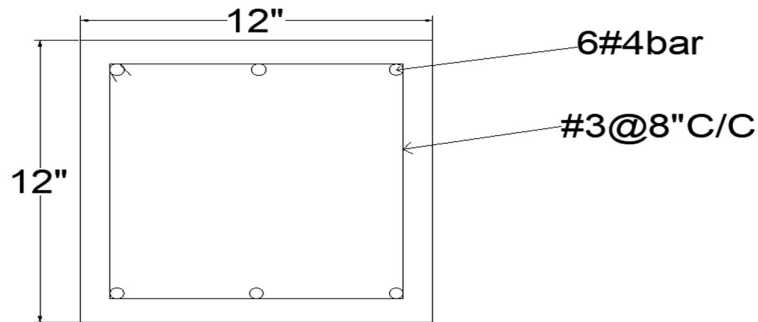
$$P_u = \phi k [0.85 f'_c (A_g - A_{st}) + A_{st} F_y]$$

Solution:

- Using area of steel to be 1% of cross sectional area
- $P_u = \phi P_n = \phi k [0.85 f'_c A_g - A_{st} + A_{st} f_y]$
- $P_u = \phi P_n = 0.65(0.8)[0.85(4)(0.99A_g) + 0.01A_g(60)]$
- $P_u = 2.06232A_g$
- $170.93 = 2.06232A_g$
- $A_g = 82.9 \text{ in}^2$
- Assuming square column 12"x12"
- For number of bars, we used #4 bar for that we needed to divide area of steel (1% of A_g) by Area of #4 bars.
- $A_{st} = 1 \text{ in}^2$
- Area of #4 bar = 0.196 in^2
- For number of bars = $A_{st} / \text{Area of \#4 bar} = 5.092$
- After rounding up, we used 6 #4 bars

Design of Ties

- 16 times longitudinal bar diameter.
 $= 16 \times 4/8 = 8$ inches
- 48 times tie diameter.
 $48 \times 3/8 = 18$ inches
- Least cross-sectional dimension 10 inches
 - We pick least of all dimensions. So we will pick spacing of ties 8"



Load Calculation (Catchment Area Method) (Interior Column)

Catchment Area = 11.886' x 16.835
Catchment area Length Slab 01 = 12.335 ft
Catchment area Length Slab 02 = 4.5 ft

Self-Weight of Slab:

$$= (150 \text{ lb/ft}^3) \times (8/12)$$
$$= 100 \text{ lb/ft}^2$$

Finishes Load:

$$= 144 \text{ lb/ft}^3 \times (3/12)$$
$$= 36 \text{ lb/ft}^2$$

Masonry Load:

$$= (144 \text{ lb/ft}^3) \times (12.67 - 2.67) \times (0.5) \times (1.886)$$
$$= 8543.52 \text{ lb}$$

Self-Weight of Beam

$$= 150 \text{ lb/ft}^3 \times (8/12) \times (32/12) \times 16.835$$
$$= 4489.33 \text{ lb}$$

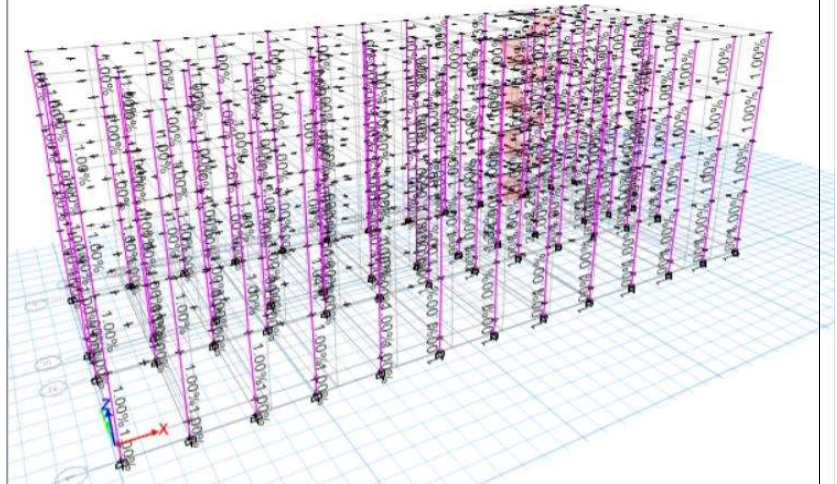
Self-Weight of Column

$$= 150 \text{ lb/ft}^3 \times (12) \times (12) \times (50.67)$$
$$= 7600.5 \text{ lb}$$

Total Dead Load:

$$= 100 + 36$$
$$= 136 \text{ lb/ft}^2$$

Point Load:



$$= 136 \text{ lb/ft}^2 \times 11.886 \text{ ft} \times 16.835 \text{ ft}$$

$$= 27213.71 \text{ lb}$$

Final Dead Load (adding masonry load and self wt of beam and column)

$$= 27213.71 + 4489.33 + 8543.52 + 7600.5$$

$$= 47861 \text{ lb or } 47.861 \text{ kips}$$

Live Load

$$\text{Live Load (classroom)} = 60 \text{ lb/ft}^2$$

$$\text{Live Load (corridor)} = 100 \text{ lb/ft}^2$$

To Point Load:

$$= (60 \text{ lb/ft}^2 \times 11.866 \text{ ft} \times 4.5 \text{ ft}) + (100 \text{ lb/ft}^2 \times 12.335 \text{ ft} \times 11.866 \text{ ft})$$

$$= 17870 \text{ lb or } 17.87 \text{ kips}$$

Ultimate Load

$$P_u = 1.2 W_o + 1.6 W_c$$

$$= (1.2 \times 47.861) + (1.6 \times 17.87)$$

$$= 86.02 \text{ kip}$$

Considering $n = 4$, G+3

$$P_u = 86.02 \times 4$$

$$P_u = 344.10 \text{ kip}$$

FOR INTERIOR COLUMN

Given:

- $\gamma_{\text{concrete}} = 150 \text{ pcf}$
- $\gamma_{\text{reinforced}} = 144 \text{ pcf}$
- Slab thickness = 8"
- Live Load for classrooms = 60 psf
- Ultimate load = $1.2P_d + 1.6P_L$
- Ultimate load = 344.10 kip

Assume:

For tied columns:

$$\Phi = 0.65$$

$$k = 0.8$$

$$F_y = 60 \text{ ksi or } 60000 \text{ psi}$$

$$F'_c = 4 \text{ ksi or } 4000 \text{ ksi}$$

Equation:

$$P_u = \phi k [0.85 f'_c (A_g - A_{st}) + A_{st} F_y]$$

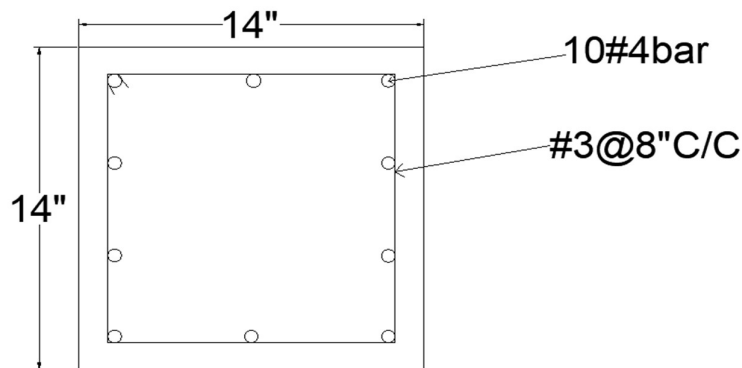
Solution:

- Using area of steel to be 1% of cross sectional area
- $P_u = \phi P_n = \phi k [0.85 f'_c A_g - A_{st} + A_{st} f_y]$
- $P_u = \phi P_n = 0.65(0.8)[0.85 f'_c(4)(0.99 A_g) + 0.01 A_g(60)]$
- $P_u = 2.06232 A_g$

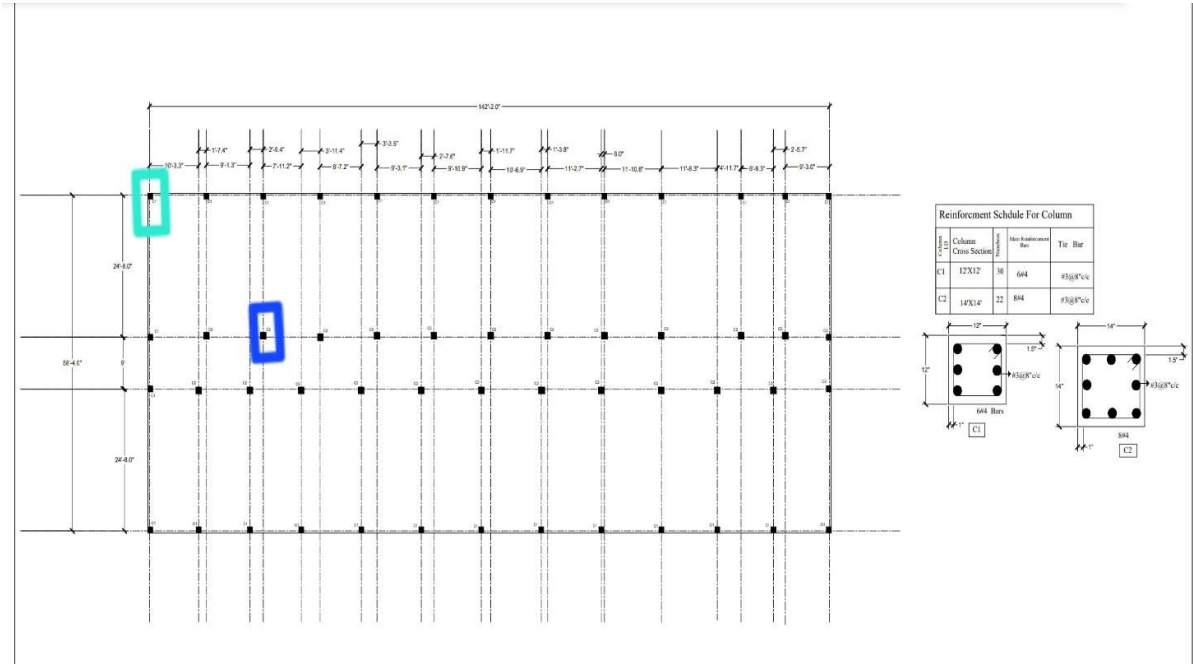
- $344.10 = 2.06232 A_g$
- $A_g = 166.85 \text{ in}^2$
- Assuming column 14"X14"
- For number of bars, we used #4 bar for that we needed to divide area of steel (1% of A_g) by Area of #4 bar.
- $A_{st} = 1.96 \text{ in}^2$
- Area of #4 bar = 0.196 in^2
- For number of bars = $A_{st} / \text{Area of \#4 bar} = 9.98$
- After rounding up, we used 10#4 bars

Design of Ties

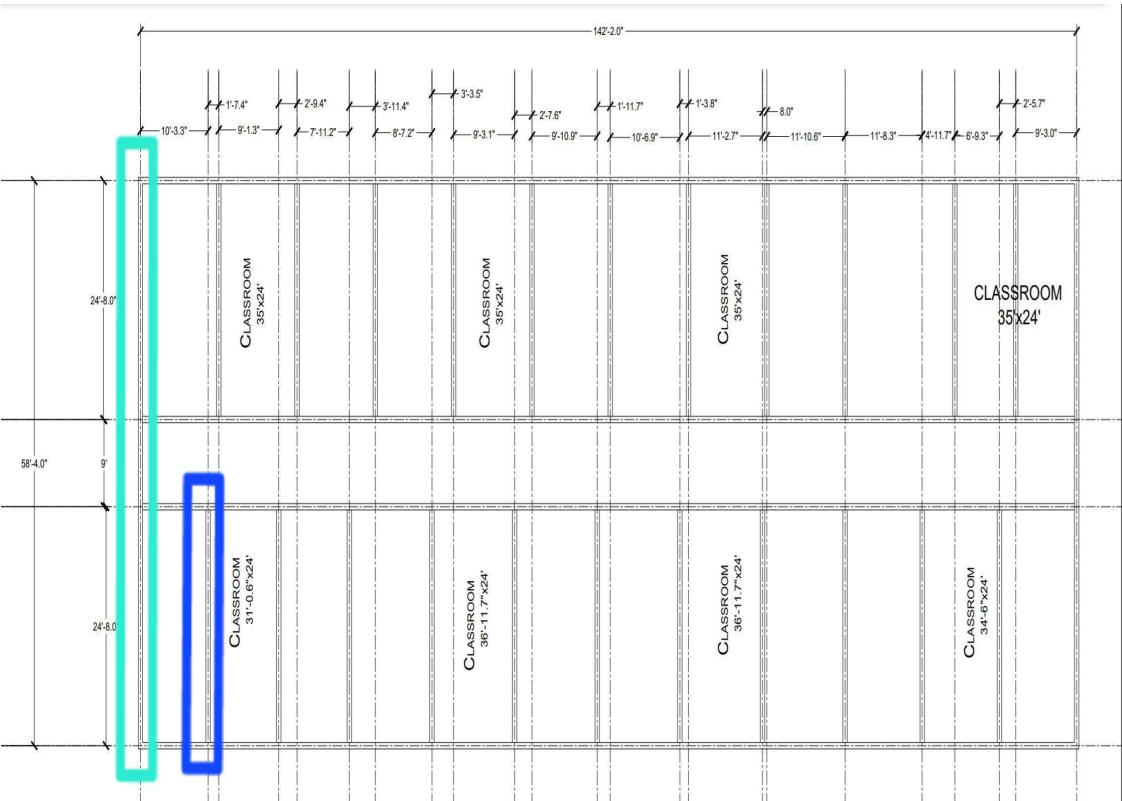
- 16 times longitudinal bar diameter.
- $= 16 * 4/8 = 8 \text{ inches}$
- 48 times tie diameter.
- $48 * 3/8 = 18 \text{ inches}$
- Least cross-sectional dimension.
- 14 inches
- We pick least of all dimensions. So we will pick spacing of ties 8"



COLUMN LAYOUT PLAN WITH DESIGNED COLUMNS



BEAM LAYOUT WITH DESIGNED BEAMS



Design of Beam:

Method:

- To design a singly reinforced beam $M_u \leq \Phi_b M_n$
- $M_u = 0.9 A_s f_y (d - a/2)$
- Where a is the depth of the stress block
- $C = T$
- $0.85 f'_c a b = A_s f_y$
- $a = \frac{A_s f_y}{0.85 f'_c b}$
- Then $M_u = 0.9 A_s f_y (d - A_s f_y / 1.7 f'_c b)$
- To ensure tension controlled section $\rho \leq \rho_{max}$
- As we know that $\rho = A_s / bd$
- So, $A_s = \rho bd$
- So the equation of M_u become $M_u = 0.9 (\rho bd) f_y [d - \frac{\rho b d f_y}{1.7 f'_c b}]$
- So, $M_u = 0.9 \rho f_y [d - \frac{\rho f_y}{1.7 f'_c}] bd^2$
- This is referred as $0.9 \rho f_y [d - \frac{\rho f_y}{1.7 f'_c}]$, R_u
- Then $M_u = R_u b d^2$

Procedure:

-

Beam Section :

Compression Factor is taken to be:

$f_y = 60$ ksi or 60000 psi

$f'_c = 4$ ksi or 4000 psi

Loads on Beam

Self-Weight of Slab = $8/12 * 150 = 100$ psf

Finishes = $3/12 * 144 = 36$ psf

Masonry = $6/12 * 144 * 10 = 720$ psf = 0.72 k/ft

Self-weight of beam = $8 * 32 / 144 * 150 = 266.667$ psf = 0.2667 k/ft

Dead Load = $136 * 11.89 / 1000 + 0.72 + 0.2667 = 2.60404$ k/ft

From Table of loads :

Class Room's Live Load lies between (50 to 70) psf

$$Live\ load = 60 * 11.89 / 1000 = 0.7134 k/ft$$

For Ultimate loads we take all loads in ksf, Formula used is ;

$$Q\ (Ultimate\ Load) = 1.2\ (Dead\ Load) + 1.6\ (Live\ Load)$$

$$Q_u = 1.2 * (2.60404) + 1.6 * (0.7134)$$

$$Q_u = 4.2296 \text{ k/ft}$$

Bending Moment & Shear Force

In order to design a beam (Of maximum strength) we consider a slab from plan consisting classroom,

TRIBUTARY AREA FOR ONE WAY SLAB

$$\text{Tributary Width} = 5.945 * 2 = 11.89 \text{ ft}$$



Washroom beam

Weight Distribution

$$\text{Ultimate Weight} = \text{Ultimate Load} * \text{Tributary Width}$$

$$W_u = Q_u * TW$$

$$W_u = 4.2296 * (5.945 + 5.945)$$

- $W_u = 5.0726 \text{ K/ft}$

$$M_u = (W_u * L^2) / 8$$

$$M_u = (4.2296 * (24)^2) / 8$$

$$M_u = 304.5312 \text{ k.ft}$$

Design for Flexural

$$M_u = R_u * b * d^2$$

$$R_u = \phi * \rho * f_y (1 - \rho * f_y / 1.7 * f'_c)$$

Assume:

$$\rho = 0.011 \%$$

$$R_u = (0.9) * (0.011) * (60) * (1 - (0.011) * (60) / (1.7) * (4))$$

- $R_u = 0.536 \text{ ksi}$

For Dimensions

$$M_u = R_u * b * d^2$$

$$M_u / R_u = b * d^2$$

$$b * d^2 = 304.5312 * 12 / 0.536$$

$$b * d^2 = 6813.451 \text{ in}^3$$

For depth of beam

$$b = 6''$$

$$b * d^2 = 6813.451$$

$$d^2 = 6813.451 / 6$$

$$d = 33.698''$$

$$b = 8''$$

$$b * d^2 = 6813.451$$

$$d^2 = 6813.451 / 8$$

$$d = 29.18''$$

For Height of Beam

$$h = d + 2.5''$$

$$h = 29.18'' + 2.5''$$

$$h = 31.68'' \text{ or } 32''$$

Area of Steel

$$\text{Area of steel} = \rho \cdot b \cdot d^2$$

$$A(s) = (0.011/100) * (8'') * (29.18'')^2$$

- $A(s) = 2.596 \text{ in}^2$
-

Size of Reinforcement

Assume # 8 bars
 $r = 8/8$ “

No. of steel bars

$$\text{No. of steel bars} = \text{Area of Steel} / \text{Area of 1 Steel Bar}$$

$$\text{No. of steel bars} = A(s) / (\pi/4 \cdot r^2)$$

$$\text{No. of steel bars} = 2.595 / (\pi/4 * (8/8)^2)$$

$$\text{No. of steel bars} = 3.305 = 4 \text{ bars}$$

$$\text{No. of steel bars} = 4$$

FOR CHECKING THE ADEQUECY

$$M_u = 304.5312 \text{ k-ft}$$

Area of Steel (Revised)

$$\text{New area of steel will be:}$$

$$A(s) = (\pi/4 \cdot r^2) * \text{No of Bars}$$

$$A(s) = (3.142/4 * (8/8)^2) * 4$$

$$A(s) = 3.141 \text{ in}^2$$

For (a)

$$a = A(s) \cdot f_y / (0.85 \cdot f'_c \cdot b)$$

$$a = (3.141 * 60) / (0.85 * 4 * 8)$$

- $a = 6.930 \text{ in}^2$

Effective Depth

$$d = 32'' - 0.5'' - (8/8)'' - (3/8)''$$

$$d = 29.1835 \text{ “}$$

\

Nominal Strength

$$\Phi. M_n = \Phi. A(s). f_y. (d - a/2)$$

$$\Phi. M_n = 0.9 * 3.141 * 60 * (29.625 - 6.930/2) / 12$$

$$\Phi. M_n = 369.828 \text{ k.ft (Nominal)}$$

Adequacy of the Beam

$$M_u \leq \Phi. M_n \text{ (Adequate)}$$

$$304.5312 \leq 369.828 \text{ (Beam is Adequate)}$$

Where,

$$\phi = 0.9 \text{ (For Tension Controlled Section)}$$

$$\rho(\max) = 0.6375 * \rho(b) \quad : \rho(b) \text{ is } \rho(\text{balanced})$$

$$\rho(\max) = 0.6375 * (0.85 * \beta_1 * (f'_c / f_y) * (87/87 + f_y)) \times 100$$

$$\rho(\max) = 0.6375 * (0.85 * 0.85 * 4/60 * (87/87 + 60)) \times 100$$

- $\rho(\max) = 1.81 \%$

$$\rho(\min) = 200 / f_y = 200 / 60000$$

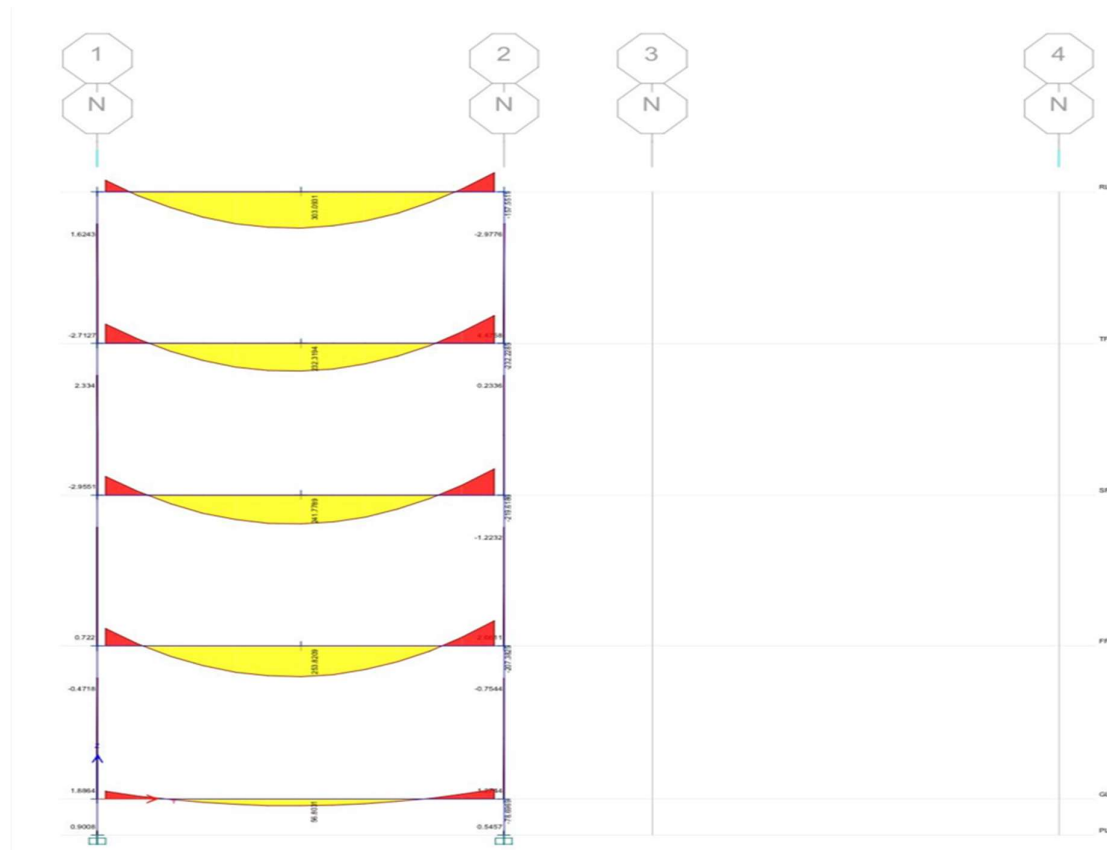
- $\rho(\min) = 0.33 \%$

Thus,

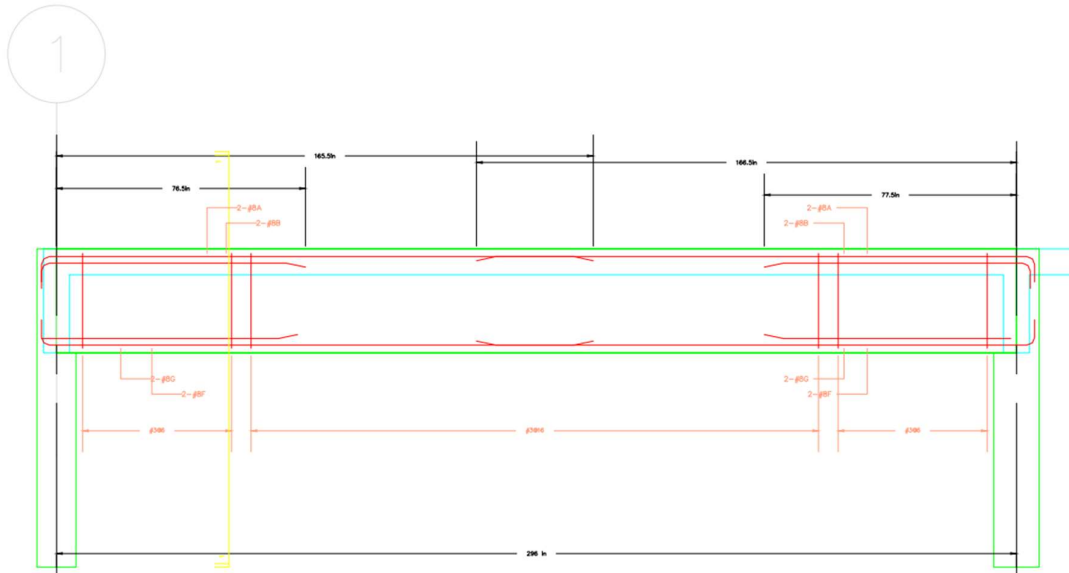
$$\rho(\min) < \rho \leq \rho(\max)$$

“It is tension controlled Section”

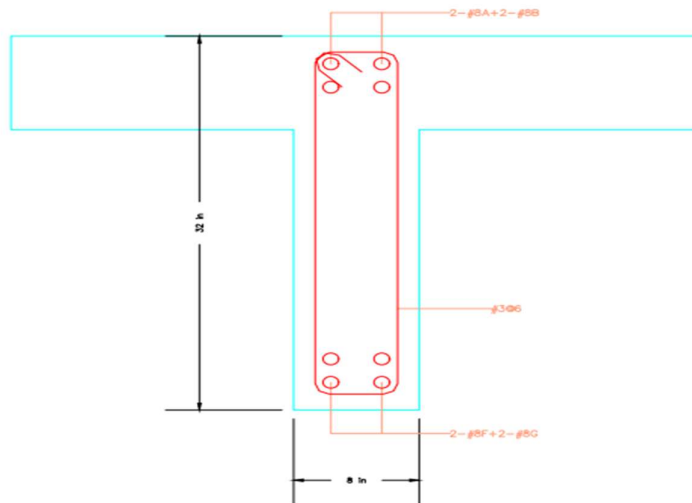
Figure:



BEAM ELEVATION:



BEAM SECTION



Shear Design :

- **Method:**

- Calculate the factored shearing force, V_u , from the applied forces acting on the structural member.
- The critical design shear value is at a section located at a distance d from the face of the support.
- Calculate ϕV_c
- If $A_v \geq A_{v1 \text{ min}}$
- $\left[8\lambda(\rho_w)^{1/3} \sqrt{f'_c} + \frac{N_u}{6Ag} \right] b_w d$
- Then calculate $\phi \lambda \sqrt{f'_c} b_w d$
- If $V_c < \phi \lambda \sqrt{f'_c} b_w d$ No shear reinforcement required
- If $\phi \lambda \sqrt{f'_c} b_w d < V_u < \phi V_c$ - Min. R/F - #3 @ max spacing
- If $V_u > \phi V_c$ - Design for Shear R/F

Spacing:

- Calculate the stirrups spacing based on the calculated $S1 = \frac{A_v f_y t d}{V_s}$
- Determine Max Spacing allowed by ACI -Smaller of S2 and S3
- $S2 = \frac{d}{2} \leq 24 \text{ in-}$ If $V_s \leq V_{c1} = 4 \sqrt{f'_c} b_w d$
- $S3 = \frac{A_v f_y t}{50 b_w}$
- Use smallest spacing to the nearest half inch For circular sections, the area used to compute V_c is the diameter times the effective depth d , where $d = 0.8$ times the diameter, ACI Code

SIMPLE SUPPORTED

- **Procedure:**

Known:

- $b=8''$
- $d=29.625''$
- $h=32''$
- Factored uniform load $= 1.2(D_L) + 1.6(L_L)$
- $UDL=4.2296$

SHEAR

$$\begin{aligned} &= 4.2296 * 24 / 2 \\ &= 50.7552k \end{aligned}$$

Find critical shear

$$50.7552/144 = x/144 - 29.625$$

$$X = 40.357k$$

$$V_c = 40.357k$$

Calculate ϕV_c

$$0.75 \{ 2 * (4000)^{1/2} + 0 \} * 8 * 29.625 / 1000$$

$$= 22.389K$$

$$\rho_w = \frac{A_s}{b_w d}$$

$$P_w = \pi / 4 * (8/8)^2 / 8 * 29.25$$

$$\rho_w = 0.00423$$

Or

$$= \{ 8 * 0.75 * (0.00423)^{1/3} * (4000)^{1/2} + 0 \} * 8 * 29.625 / 1000$$

$$= 14.48k$$

Since V critical is greater shear reinforcement is required

Calculate V_{c1} and V_{c2}

$$V_{c1} = 4 \sqrt{f'c} \quad b_w d$$

$$V_{c1} = 4 (4000)^{1/2} * 8 * 29.25 / 1000$$

$$V_{c1} = 59.703 \text{ kips and } V_{c2} = 119.407 \text{ kips}$$

SPACING

Assume #3 bar – Area = 0.11 in²
 $A_v = 0.22 \text{ in}^2$ for 2 legs

S1:

$$\text{For } S1 = \frac{A_v f_y t d}{V_s}$$

$$V_s = V_u - V_c / \phi$$

$$V_s = (40.432 - 22.38) / 0.75$$

$$V_s = 24.05 \text{ k}$$

$$f_y t = 60 \text{ ksi}$$

$$d = 29.625 \text{ in}$$

$$S1 = 16.319 \text{ in}$$

$$S2 = d/2 \leq 24 \text{ “}$$

$$S2 = 29.625/2 = 14.750 \text{”}$$

$$S2 = 14.750 \text{ in}$$

S3

$$S3 = \frac{A_v f_y t}{50 b_w}$$

$$S3 = 0.22 * 60000 / 50 * 8$$

$$S3 = 33.134 \text{ in}$$

$$S4 = 34.926$$

USE SPACING 14.750

Selection of v_s check

$$A_{v,min} = 0.75 * (f'c)^{1/2} * (b * s / f_y t)$$

$$= 0.75 * (4000)^{1/2} * (8 * 14.750 / 60000)$$

$$A_{v,min}=0.093 \text{ in}^2$$

$$A_{v,min}=50*b*s/f_y$$

$$A_{v,min}=50*8*14.750/60000$$

$$A_{v,min}=0.098$$

CONTINUOUS BEAM DESIGN

Procedure:

MAX MOMENT FROM ETABS = 220k-ft

$$M_u = R_u * b * d^2$$

$$R_u = \phi * \rho * f_y (1 - \rho * f_y / 1.7 * f'_c)$$

Assume:

$$\rho = 0.011 \%$$

$$R_u = (0.9) * (0.011) * (60) * (1 - (0.011) * (60) / (1.7) * (4))$$

- $R_u = 0.536 \text{ ksi}$

For Dimensions

$$M_u = R_u * b * d^2$$

$$M_u / R_u = b * d^2$$

$$b * d^2 = 220 * 12 / 0.536$$

$$b * d^2 = 4925.3731 \text{ in}^3$$

For depth of beam we assume base = 8"

$$b = 8"$$

$$b * d^2 = 4925.3731 \text{ in}^3$$

$$d^2 = 4925.3731 / 8$$

$$d = 24.812"$$

$$d = 29.5"$$

For Height of Beam

$$h = d + 2.5"$$

$$h = 29.5" + 2.5"$$

$$h = 32"$$

Area of Steel

$$\text{Area of steel} = \rho * b * d$$

$$A(s) = (0.011) * (8") * (29.5)$$

- $A(s) = 2.596 \text{ in}^2$

Size of Reinforcement

Assume # 9 bars

$$r = 9/8 "$$

No. of steel bars

$$\text{No. of steel bars} = \text{Area of Steel} / \text{Area of 1 Steel Bar}$$

$$\text{No. of steel bars} = A(s) / (\pi/4 * r^2)$$

$$\text{No. of steel bars} = 2.596 / (\pi/4 * (9/8)^2)$$

$$\text{No. of steel bars} = 2.6116 = 4 \text{ bars}$$

$$\text{No. of steel bars} = 4$$

FOR CHECKING THE ADEQUECY

$$M_u = 220 \text{ k-ft}$$

For (a)

$$a = A(s) \cdot N_b \cdot f_y / (0.85 \cdot f'_c \cdot b)$$

$$a = (2.596 \cdot 4 \cdot 60) / (0.85 \cdot 4 \cdot 8)$$

- $a = 22.9058 \text{ in}^2$

Effective Depth

$$d = 29.5'' - (3/8) - (4.5/8) - 1.5$$

$$d = 27.0625''$$

Nominal Strength

$$\Phi \cdot M_n = \Phi \cdot A(s) \cdot f_y \cdot (d - a/2)$$

$$\Phi \cdot M_n = 0.9 \cdot 2.596 \cdot 4 \cdot 60 \cdot (27.0625 - 22.9058/2) / 12$$

$$\Phi \cdot M_n = 729.4053 \text{ k.ft (Nominal)}$$

Adequacy of the Beam

$$M_u \leq \Phi \cdot M_n \text{ (Adequate)}$$

$$220 \text{ k-ft} \leq 729.4053 \text{ k-ft (X- Section is Adequate)}$$

Where,

$$\phi = 0.9 \text{ (For Tension Controlled Section)}$$

$$\rho(\max) = 0.6375 \cdot \rho(b) \quad : \rho(b) \text{ is } \rho(\text{balanced})$$

$$\rho(\max) = 0.6375 \cdot (0.85 \cdot \beta_1 \cdot (f'_c / f_y) \cdot (87/87 + f_y)) \cdot 100$$

$$\rho(\max) = 0.6375 \cdot (0.85 \cdot 0.85 \cdot 4/60 \cdot (87/87 + 60)) \cdot 100$$

- $\rho(\max) = 1.807 \%$

$$\rho(\min) = 200 / f_y = 200 / 60000$$

- $\rho(\min) = 0.33 \%$

Thus,

$$\rho(\min) < \rho \leq \rho(\max)$$

“It is tension controlled Section”

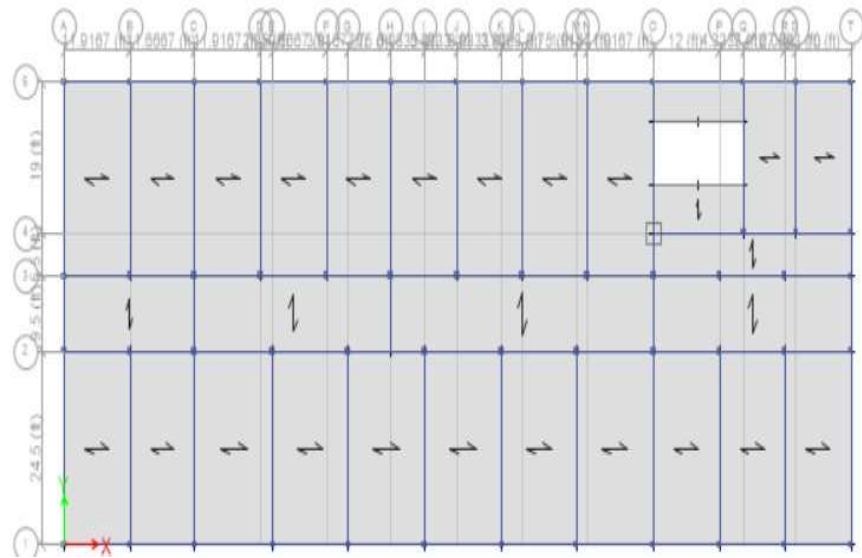
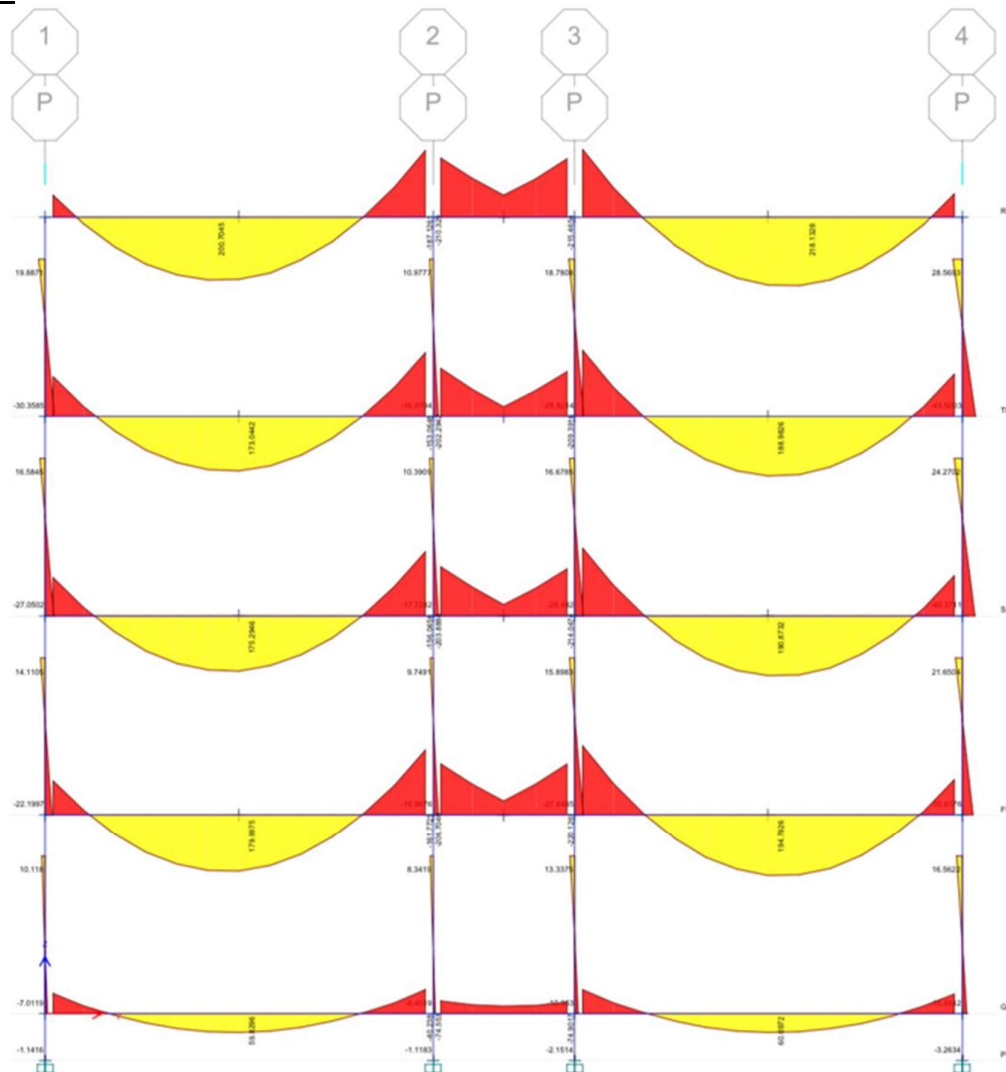
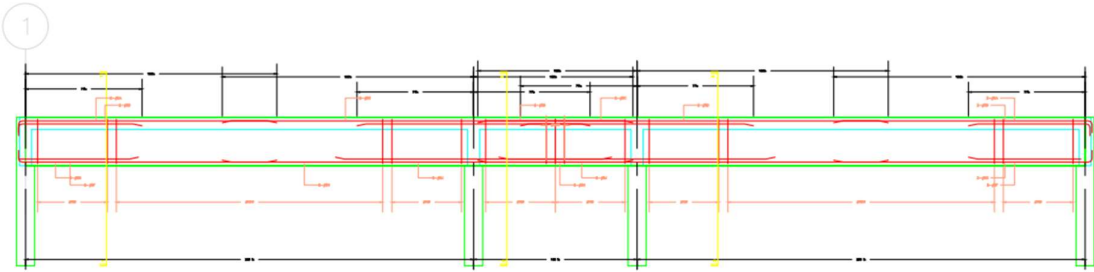


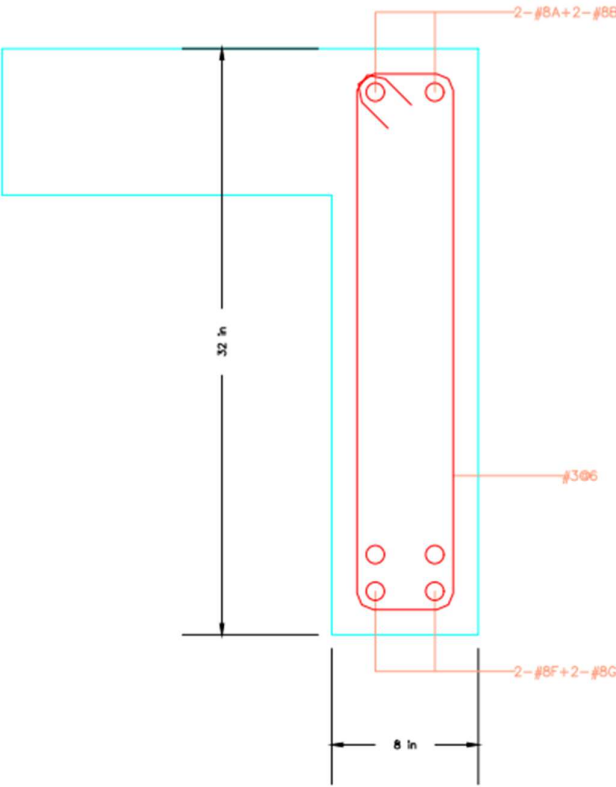
Figure:



BEAM ELEVATION



BEAM SECTION



Shear For Continuous Beam:
Maximum Shear From ETABS=74.05kips

$$V_{max}=74.05$$

Using Similar Triangle:

$$74.05/144'' = V_u / (144''-29.5'')$$

$$V_u = 58.880 \text{ kips (Critical Shear)}$$

Calculate ϕV_c

$$= 0.75 \{ 2 * (4000)^{1/2} + 0 \} * 8 * 29.5 / 1000$$
$$= 22.38 \text{ kips}$$

$$\rho_w = A_s / b_w * d$$

$$P_w = \pi / 4 * (8/8)^2 / 8 * 29.25$$

$$\rho_w = 0.00423$$

Or

$$= \{ 8 * 0.75 * (0.00423)^{1/3} * (4000)^{1/2} + 0 \} * 8 * 29.5 / 1000$$
$$= 14.48 \text{ k}$$

V_u is greater than ϕV_c
Shear is reinforced

Calculate $\phi V_c / 2$

$$\Phi V_c / 2 = 22.38 / 2$$

$$\Phi V_c / 2 = 11.19 \text{ kips}$$

Calculate V_{c1} and V_{c2}

$$V_{c1} = 4 (f'c)^{1/2} b_w d$$

$$V_{c1} = 4 (4000)^{1/2} * 8 * 29.5 / 1000$$

$$V_{c1} = 59.70 \text{ kips}$$

$$V_{c2} = 8 (f'c)^{1/2} b_w d$$

$$V_{c2} = 8 (4000)^{1/2} * 8 * 29.5 / 1000$$

$$V_{c2} = 119.407 \text{ kips}$$

Calculate V_s

$$V_s = V_u - V_c / \Phi$$

$$V_s = 58.880 - 22.38 / 0.75$$

$$V_s = 48.667$$

$$\underline{V_s < V_{c2}} \quad (\text{No Section Revision})$$

Calculate Spacing

For S1

$$S1 = A_v f_y t * d / V_s$$

$$S1 = 0.22 * 60 * 29.5 / 48.667$$

$$S1 = 8''$$

For S2

$$S2 = 29.5 / 2$$

$$S2 = 14''$$

For S3

$$S3 = 0.22 * 60000 / 0.75 * (4000)^{1/2} * 8$$

$$S3 = 33''$$

For S4

$$S4 = 0.22 * 60000 / 0.75 * (4000)^{1/2} * 8$$

$$S4 = 24.78''$$

Now

$$\Phi V_n = \Phi V_c + \Phi V_s$$

For ΦV_s

$$S1 = A_v f_{yt} * d / V_s$$

Or

$$V_s = A_v f_{yt} * d / S1$$

$$V_s = 0.22 * 60 * 29.5 / 16''$$

$$V_s = 24.3375$$

$$\Phi V_s = 0.75 * 24.3375$$

$$\Phi V_s = 18.2531$$

$$\Phi V_n = \Phi V_c + \Phi V_s$$

$$\Phi V_n = 22.38 + 18.2531$$

$$\Phi V_n = 40.6331$$

Using Similar Triangle

$$74.05 / 144 = 40.6331 / X$$

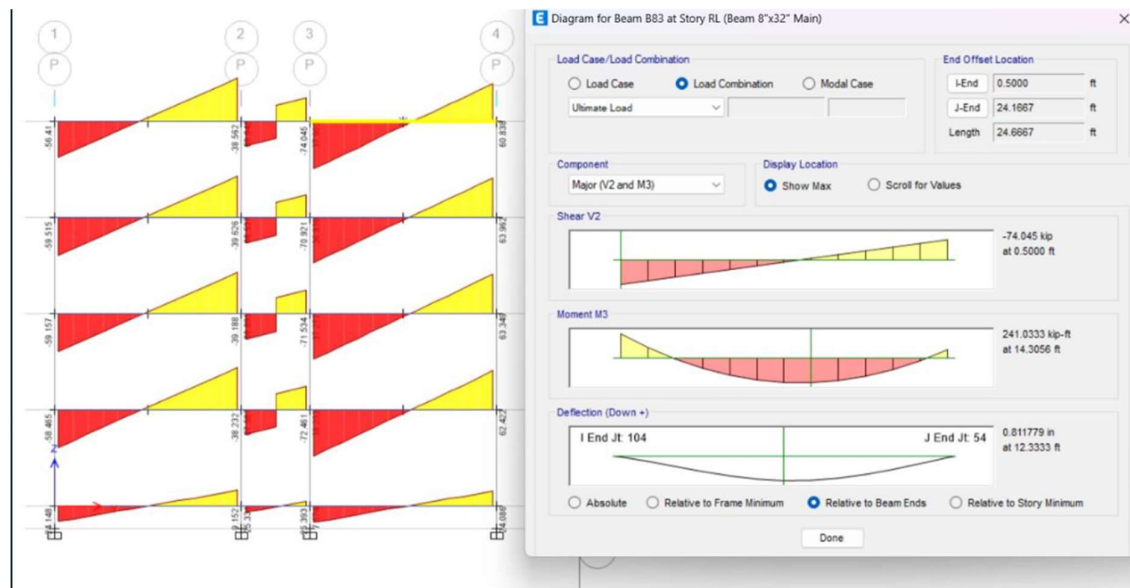
$$X = 79.016''$$

Similarly

$$74.05 / 144 = 11.19 / X (\text{No shear})$$

$$X (\text{no shear}) = 21.7604''$$

Figure



SOLUTION BY EXCEL:

REINFORCED CONCRETE DESIGN - I														
COMPLEX ENGINEERING PROBLEM (COLUMN DESIGN)														
DATA			Green Colours are Input Dark and Light Blue are Output											
Specific weight of RCC	pcf	150												
Specific weight of PCC	pcf	144												
Slab thickness	inches	8												
Width of beams	inches	8												
Depth of beams	inches	12												
Thickness of columns	inches	12												
Width of columns	inches	12												
Height of columns	ft	50.97												
Height of wall	ft	10												
Thickness of wall	inches	6												
Reduction factor ϕ		0.65												
f_y		60												
strength of steel f_y	ksi	60												
strength of concrete f_c	ksi	4												
Self weight of Wall	Ru/ft	720	Dead Load (stairs)	pcf	63									
Self weight of finishes	pcf	36	Live Load (stairs)	pcf	100									
Self weight of Slab	pcf	100												
Self weight of Beams	Ru/ft	246.667												
Self weight of Column	lbs	7800.5												

Slab 01

C2

Slab 02

Slab 01

C2

Slab 02

DESIGN OF COLUMN:

DESIGN OF CORNER & EDGE COLUMN

Live Load (Class rooms)

pcf

60

Catchment area Length

ft

22.500

Live Load (Corridor)

pcf

100

Catchment area Width

ft

5.945

Live Load (Roof)

pcf

0

Dead Load of Slab

lb/ft

7800.5

Live Load of Slab + Roof

lb/ft

17899.7

Dead Load of Beams

lb/ft

246.60

Dead Load of Wall

lb/ft

8881.2

Dead Load of Finishes

lb/ft

2539.8

Dead Load of Stairs

lb/ft

6789.2

Dead Load of Column

lb/ft

7800.5

Total Live Load

lb/ft

45999.0

Total Dead Load

lb/ft

25744.5

Ultimate Load (Pu)

lb/ft

136933.5

Thickness of Column (h)

inches

30

Width of Column (b)

inches

30

Design of Ties

Diameter of Long Bars

inches

6

Diameter of Ties

inches

6

Least cross section dim.

inches

10.0

Adequacy Check for Column

Pu (kip)

190.1133

Phi (kip)

206.232

Pu < Phi

Column is Adequate

Assume 1%

Ratio = As/Ag

As < 0.01 Ag

AG

92.00354

93

Dimension

9.150935

10

Assume 9.4 bars

9.339345045

9

AG

930

10

Dimension

9.150935

10

No of Bars

9.339345045

9

No of Bars

9

10

Assume 9.3 Ties

9.339345045

9

Selected Spacing

9.339345045

9.3

8.0

Green Colours are Input

Dark and Light Blue Colours are Output

DESIGN OF INTERIOR COLUMN

Live Load (Class rooms)

pcf

60

Catchment area Width

ft

11.886

Live Load (Corridor)

pcf

100

Catchment area Length/Slab 01

ft

5.312

Live Load (Roof)

pcf

0

Catchment area Length/Slab 02

ft

6.5

Catchment area Total Length

ft

10.875

Dead Load of Slab

lb/ft

20020.7

Live Load of Slab + Roof

lb/ft

17899.7

Dead Load of Beams

lb/ft

4488.0

Dead Load of Wall

lb/ft

9327.9

Dead Load of Finishes

lb/ft

17601.8

Dead Load of Stairs

lb/ft

13606.6

Dead Load of Column

lb/ft

7800.5

Total Live Load

lb/ft

157480

Total Dead Load

lb/ft

67399.6

Ultimate Load (Pu)

lb/ft

158497

AG

1085.0545

107

Dimension

12.3233889

13

AG

136

14

Dimension

12.3233889

13

Assume 9.4 bars

9.339345045

9

AG

930

10

Dimension

9.150935

10

No of Bars

9.339345045

9

No of Bars

9

10

Assume 9.3 Ties

9.339345045

9

Selected Spacing

9.339345045

9.3

8.0

Design of Ties

Diameter of Long Bars

inches

6

Diameter of Ties

inches

6

Least cross section dim.

inches

14.0

Adequacy Check for Column

Pu (kip)

344.10022

Phi (kip)

346.71822

Pu < Phi

Column is Adequate

DESIGN OF SIMPLY SUPPORTED BEAM:

DESIGN OF BEAM			
Total Load	k/ft	0.72	
Live Load (Class rooms)	psf	60	IBC-2018
Live Load (Corridors)	psf	100	IBC-2018
Self weight of beam	k/ft	0.266666667	
Dead Load	psf	136	
Tributary Width (One Slab)	ft	11.75	
Total Live Load	k/ft	0.705	
Total Dead Load	k/ft	2.584666667	
Ultimate Load	k/ft	4.2296	
Maximum Moment (Mu)	k-ft	304.5312	
Effective Depth of Beam	in	29.5	
Mu = Ru bd ² /4			
Moment of resistance (Ru)	kai	0.525	
Maximum % of Steel			
For Tension Controlled Section			
Rou (max) = 0.6375 Rou(b)	Beta 1	0.85	
For Rou (b)			
Rou (b)	0.0285	2.851%	
Rou (max)	0.0181	1.807%	
Minimum % of Steel			
Rou (min) = 200/fy			
Rou (min)	0.003	0.33%	
(0.33% - 2.851%)			
Area of Steel			
Area of Steel (As)	sq. in	2.586	
No. of bars = As/Ab			
Assuming #8bars			
Ab	0.785398163		
No. of bars (n)		3.308	4.0
CHECKING ADEQUACY			
Ultimate Load	k/ft	4.2296	d 29.625
Maximum Moment (Mu)	k-ft	304.5312	a 6.930
Design Strength (ft Min)	k-ft	369.828	
304.5312	369.828	X-section is adequate	
		(Self Weight of beam is included)	

DESIGN OF SHEAR IN RCC BEAM			
Live Load (Class rooms)	psf	60	IBC-2018
Live Load (Corridors)	psf	100	IBC-2018
Self weight of beam	lb/ft	0.266666667	
Depth of Beam	inches	32	
Width of Beam	inches	8	
Effective depth (d)	inches	29.5	
Total Live Load	k/ft	0.705	
Total Dead Load	k/ft	2.584666667	
Ultimate Load	k/ft	4.2296	
(Vu)	kips	50.7552	
Critical Shear			
Critical Shear (Vcritical)	kips	40.357	
Calculate ft Vc			
OPTION 1			
ft (Vc)	kips	22.389	
OPTION 2			
ft (Vc)	kips	21.235	
USE		22.389	
Shear Requirement			
Shear Rf Needed			

FOR BEAM			
SPAN LENGTH OF BEAM (ft)	24		
Single Layer R/F (in)	2.5		
Double Layer R/F (in)	3.5		
ft	0.75		
SPAN LENGTH OF BEAM (inch)	144		
Normal Wt. Concrete (Lambda)	1		
Factor (Nou/s*Ag)	0		
Rou (w)	0.0		
As	3.1		
Ag			
Vc1			
Critical Shear (Vcritical)	kips	59.704	
Vc2			
Critical Shear (Vcritical)	kips	119.408	

Calculate $f_t V_c/2$			
$f_t (V_c/2)$	Kips	11.194	
Calculate $f_t V_{c1}$			
$f_t (V_{c1})$	Kips	59.704	
Calculate $f_t V_{c2}$			
$f_t (V_{c2})$	Kips	119.408	
Calculate V_s			
V_s	Kips	23.958	
Section Revision Req			
No Section Revision			
SPACING			
S1	in	16.319	Assume # 3 bar
S2	in	14.750	Area of Bar
S3	in	33.134	FOR TWO LEGS
S4	in	34.926	
USE SPACING			14.750
AREA MIN			
A_v, min	in^2	0.093	A_v, min
Selection For V_s Check			
A_v, min	in^2	0.098	in
Since			
Check is Ok we can use Either Of the Formula To Calculate V_c			

DESIGN OF CONTINUOUS BEAM:

DESIGN OF BEAM				Continuous Beam	
Maximum Moment (Mu)		k.ft	220	Maximum Moment From Etabs	
Maximum % of Steel					
For Tension Controlled Section					
Rou (max) = 0.63375 Rou(b)		Beta 1		0.85	
For Rou (b)					
Rou (b)		0.0285	2.851%		
Rou (max)		0.0181	1.807%		
Minimum % of Steel				% of Steel Required	
Rou (min) = 200/fy				Rou (req) = 0.85 f'c/fy [1-sqrt(1-(2Ru/(0.85+0.85 f'c)))]	
Rou (min)	0.003	0.33%	Rou (req)		0.0107378 1.073778
(0.33% - 2.851%)					
Area of Steel		Assuming Rou		0.011	
Dimension Design of Beam					
Ru=pi*row*fy*(1-(row*fy/(1.7*f'c)))		phi=		0.9	
Ru	Ksi	0.536			
Mu = Ru bd^2					
b*d^2=Mu/Ru					
b*d^2=	4922.186 inch^3				
b=	6 inch	d=	28.64200304 inch		
b=	8 inch	d=	24.80470225 inch		
b=	8 inch				
d=	24.80470225 inch				
h=d*2.5	27.30470225 inch				
Cross Section of Beam 1 is=					
Width	b=	8 inch			
Height	h=	32 inch			
Effective Depth	d=	29.5			
Area of Steel					
Area of Steel (As)	sq. in	2.596			
No. of bars = As/Ab		ASSUMING #9bars			
		Ab		0.9940196	
No. of bars (n)	2.612		3		
CHECKING ADEQUACY					
				d	29.5625
Maximum Moment (Mu)	k.ft	220		a	17.179
Design Strength (fi Mn)	k.ft	735.013			
220	735.013	X-section is adequate			
				(Self Weight of beam is included)	

Ultimate Loads of Columns through Etabs

Column	Output Case	P
		kip
C1	Ultimate Load	-107.6
C2	Ultimate Load	-175.8
C3	Ultimate Load	-180.2
C4	Ultimate Load	-183.96
C5	Ultimate Load	-181.96
C6	Ultimate Load	-175.75
C7	Ultimate Load	-180.69
C8	Ultimate Load	-183.13
C9	Ultimate Load	-181.31
C10	Ultimate Load	-182.2
C11	Ultimate Load	-198.39
C12	Ultimate Load	-138.26
C13	Ultimate Load	-73.393
C14	Ultimate Load	-107.69
C15	Ultimate Load	-175.78
C16	Ultimate Load	-195.95
C17	Ultimate Load	-212.7
C18	Ultimate Load	-209.61
C19	Ultimate Load	-211
C20	Ultimate Load	-210.05
C21	Ultimate Load	-209.59
C22	Ultimate Load	-192.98
C23	Ultimate Load	-184.19
C24	Ultimate Load	-183.55
C25	Ultimate Load	-106.9
C26	Ultimate Load	-163.07
C27	Ultimate Load	-163.05
C28	Ultimate Load	-97.475
C29	Ultimate Load	-164.18
C30	Ultimate Load	-265.37
C31	Ultimate Load	-261.74
C32	Ultimate Load	-276.39
C33	Ultimate Load	-298.18
C34	Ultimate Load	-262.65
C35	Ultimate Load	-302.15
C36	Ultimate Load	-281.33
C37	Ultimate Load	-276.64
C38	Ultimate Load	-290.74
C39	Ultimate Load	-266.89
C40	Ultimate Load	-309.04
C41	Ultimate Load	-282.46
C42	Ultimate Load	-263.51
C43	Ultimate Load	-298.99
C44	Ultimate Load	-262.42
C45	Ultimate Load	-261.76
C46	Ultimate Load	-297.36
C49	Ultimate Load	-169.95
C50	Ultimate Load	-262.12
C51	Ultimate Load	-260.43
C52	Ultimate Load	-266.99

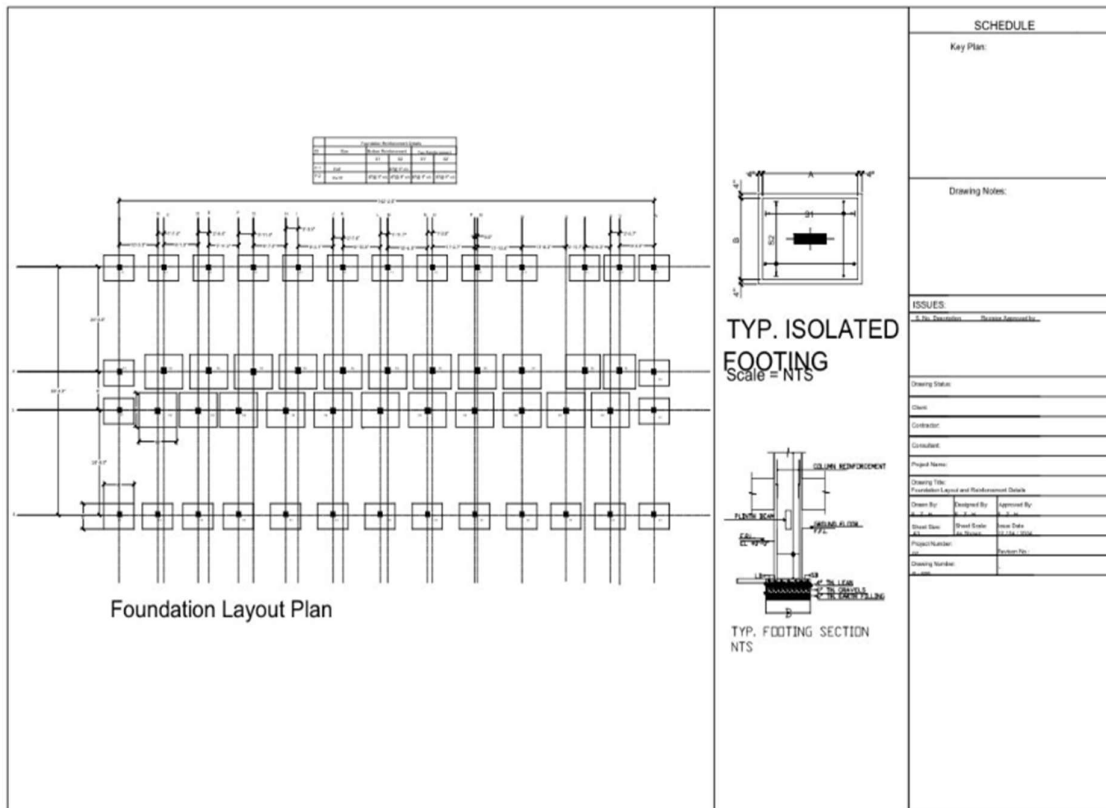
Footing Design: **METHOD**

- Assume total depth of footing
- Calculate the effective soil pressure
Weight Footing = $\gamma_{\text{concrete}} \times \text{depth of footing}$
- Calculate Area of Footing
Area of Footing = $\text{Service load} / \text{Effective soil pressure}(q_a)$
- Calculate Ultimate load
 $P_u = 1.2 D + 1.6 L$
- Net Upward pressure
 $q_u = P_u / \text{Area}$
- Check for one-way shear
The critical section is at a distance d from the face of the column. In the Longitudinal direction
 $V_{u1} = (L/2 - c/2 - h) * q_u * b$
 $\phi V_c = \phi [2\lambda \sqrt{f_c} / 1000 + N_u / 6A_g] * b_w * d$
 $d = V_{u1} / 4\phi \lambda \sqrt{f_c} * b_w$
- Check for Two-way shear
The critical section is at a distance $d/2$ from the face of the column on 4 sides
The shearing force V_{u2} for square column $b_0 = 4(c + d)$
The shearing force V_{u2} for rectangle column $b_0 = 2[(c_1 + d) + (c_2 + d)]$
The section is subjected to a vertical downward load, P_u , and a vertical upward pressure, q_u Therefore,
 $d = V_{u2} / 4\phi \lambda \sqrt{f_c} b_0$
 $b_0 = 4c + d_{\text{effective}}$
 $V_{u2} = P_u - q_u(c + h)^2$
 $\beta = \text{longer} / \text{shorter}$
 $d_1 = V_{u2} / 4\phi \lambda \sqrt{f_c} b_w$
 $d_2 = V_{u2} / \phi (a_s * d_{\text{effective}} / b_0 + 2) \lambda s \lambda \sqrt{f_c} b_0$
- Design steel reinforcement in the longitudinal direction.
The critical section is a face of supports
($L/2 - C/2$)
 $= q_u * (L/2 - C/2)^2 * b$
 $R_u = M_u / b d^2$
 $A_s = \rho b d$
Min $A_s_{\text{shrinkage}} = 0.0018 b h$
Min $A_s_{\text{Flexure}} = 0.0033 b h$
- For Shorter Side
Required Area per foot = $A_s / \text{Longer side}$

Spacing = $A_b / \text{Area per foot} * 12$

- Check the bearing stress at the base of the column

Foundation Layout Plan:



Procedure: **Interior Footing Design**

$$P \text{ (Service dead)} = 50.6 \text{ kips}$$

$$P \text{ (Service Live)} = 17.87 \text{ kips}$$

- Assume total depth of footing
Assume total depth of footing = 2ft

- Calculate the effective soil pressure
Weight Footing = $\gamma_{\text{concrete}} \times \text{depth of footing}$
Self-weight of footing = $7 \times 7 \times 150 \times 2 / 1000 = 14.7 \text{ kips}$

$$P(\text{Service dead}) = 50.6 + 14.7 = 65.3 \text{ kips}$$

$$P(\text{Service}) = (65.3 + 17.87) \times 4 = 332.68 \text{ kips}$$

1 Calculate Ultimate load

$$P_u = 1.2 D + 1.6 L$$

$$P_u = [1.2(65.3) + 1.6(17.87)] \times 4$$

$$P_u = 427.808 \text{ kips}$$

- $q_a = 5 \text{ ksf}$
- Column size = 14"x14"

2 Assume Dimensions of Footing

$$7' \times 7' \times 2'$$

3 Effective Soil Pressure

$$q_f = 150 \times 2 = 300 \text{ psf}$$

$$q(\text{soil}) = 100 \times 2 = 200 \text{ psf}$$

$$q(\text{total}) = 300 + 200 = 500 \text{ psf}$$

$$q(\text{effective}) = 5000 - 500$$

$$q(\text{effective}) = 4500 \text{ psf}$$

4 Calculate Area of Footing

$$A = P_s / q(\text{eff})$$

$$A = 332.68 / 4.5$$

$$A = 73.92 \text{ ft}^2$$

$$\text{Assume one side} = 8'$$

$$\text{So, } 73.935 / 8 = 9.24' = 10'$$

$$\text{So, our dimensions are } 8' \times 10'$$

5 Ultimate Pressure

$$\text{Effective soil pressure} = q_u = 427.808 / 8 \times 10$$

$$\text{Effective soil pressure} = q_u = 5.348 \text{ ksf}$$

6 Check for One-Way Shear

The critical section is at a distance d from the face of the column. In the longitudinal 10 ft. direction

$$V_{u1} = (L/2 - C/2 - d) \times q_u \times b$$

$$V_{u1} = (10/2 - 1.167/2 - 19.5/12) \times 5.348 \times 8$$

$$V_{u1} = 119.43 \text{ kips}$$

$$d = V_{u1} / 2\phi \lambda \sqrt{f_c} \times b_w$$

$$d = 119.43 / 2 \times 0.75 \times 1 \times \sqrt{4000} \times 8.5 \times 12$$

$$d = 15.736'$$

So,

$$19.5'' > 15.736'' \text{ OKAY}$$

7 Check for Two-Way Shear

Square column

$$b_o = 4(c+d)$$

$$b_o = 4(14+19.5)$$

$$b_o = 134 \text{ inch}$$

$$V_{u2} = P_u - q_u(c+d)^2$$

$$V_{u2} = 427.808 - 5.34(1.167+19.5/12)^2$$

$$V_{u2} = 386.181 \text{ kips}$$

$$\beta = \text{longer/shorter}$$

$$\beta = 10/8$$

$$\beta = 1.25 < 2$$

$$d_1 = V_{u2} / 4\phi \lambda \sqrt{f_c} * b_o$$

$$d_1 = 386.181 \times 1000 / 4 \times 0.75 \times 1 \times \sqrt{4000} \times 134$$

$$d_1 = 15.189''$$

$$d_2 = V_{u2} / \phi (a_s * \text{effective} / b_o + 2) \lambda_s \lambda \sqrt{f_c} b_o$$

$$\text{Where, } \lambda_s = 1, a_s = 40$$

$$d_2 = 386.181 \times 1000 / 0.75 (40 \times 19.5 / 150 + 2) \times 1 \times 1 \times \sqrt{4000} \times 134$$

$$d_2 = 8.438 \text{ in}$$

8 Design Steel Reinforcement In The Longitudinal Direction

$$M_u = q_u/2(L/2 - C/2)^2 * b$$

$$M_u = 5.34/2(10/2 - 1.167/2)^2 * 8$$

$$M_u = 417.26 \text{ k-ft}$$

$$R_u = M_u / b d^2$$

$$R_u = 417.26 \times 1000 \times 12 / 8 \times 12 \times 19.5^2$$

$$R_u = 137.166 \text{ psi}$$

$$\rho = 0.85 \times 4 / 60 [1 - \sqrt{1 - 2 \times 137.166 / (0.85)(0.9)(4000)}]$$

$$\rho = 0.003$$

$$A_s = \rho b d$$

$$A_s = 0.003 \times 8 \times 12 \times 19.5$$

$$A_s = 5.616 \text{ in}^2$$

$$\text{Min } A_s \text{ shrinkage} = 0.0018 b h$$

$$\text{Min } A_s \text{ shrinkage} = 0.0018 \times 8 \times 12 \times 24$$

$$\text{Min } A_s \text{ shrinkage} = 4.4 \text{ in}^2$$

$$\text{Min } A_s \text{ Flexure} = 0.0033 b d$$

$$\text{Min } A_s \text{ Flexure} = 0.0033 \times 8 \times 12 \times 19.5$$

$$\text{Min } A_s \text{ Flexure} = 6.24 \text{ (SELECTED)}$$

For Shorter Side

$$\text{Required Area per foot} = A_s / \text{Longer side}$$

$$\text{Required Area per foot} = 6.24 / 8$$

$$\text{Required Area per foot} = 0.78 \text{ in}^2 / \text{ft}$$

Spacing

Using # 7 bar

$$\text{Spacing} = A_b / \text{Area per foot} * 12$$

$$\text{Spacing} = 0.6013 / 0.78 * 12$$

$$\text{Spacing} = 9.251$$

$$\text{Spacing} = 9'' \text{ C/C}$$

Corner And Edge Footing Design

- Assume total depth of footing

$$\text{Assume total depth of footing} = 2\text{ft}$$

- Calculate the effective soil pressure

$$\text{Weight Footing} = \gamma_{\text{concrete}} \times \text{depth of footing}$$

$$\text{Self-weight of footing} = 7 \times 7 \times 150 \times 2 / 1000 = 14.7 \text{ kips}$$

$$P(\text{Service dead}) = 32.488 + 14.7 = 47.1888 \text{ kips}$$

$$P(\text{Service}) = (47.188 + 4.399) \times 4 = 206.355 \text{ kips}$$

- 1 Calculate Ultimate load

$$P_u = 1.2 D + 1.6 L$$

$$P_u = [1.2(47.1888) + 1.6(4.399)] \times 4$$

$$P_u = 254.665 \text{ kips}$$

- $q_a = 5 \text{ ksf}$
- Column size = 14''x14''

- 2 Assume Dimensions of Footing

$$7' \times 7' \times 2'$$

- 3 Effective Soil Pressure

$$q_f = 150 \times 2 = 300 \text{ psf}$$

$$q(\text{soil}) = 100 \times 2 = 200 \text{ psf}$$

$$q(\text{total}) = 300 + 200 = 500 \text{ psf}$$

$$q(\text{effective}) = 5000 - 500$$

$$q(\text{effective}) = 4500 \text{ psf}$$

- 4 Calculate Area of Footing

$$A = P_s / q(\text{eff})$$

$$A = 206.355 / 4.5$$

$$A = 45.857 \text{ ft}^2$$

$$\text{Assume one side} = 8'$$

$$\text{So, } 45.857 / 8 = 5.73' = 6'$$

So, our dimensions are 6'x8'

5 Ultimate Pressure

$$\text{Effective soil pressure} = q_u = 254.655 / 6 \times 8$$

$$\text{Effective soil pressure} = q_u = 5.3055 \text{ ksf}$$

6 Check for One-Way Shear

The critical section is at a distance d from the face of the column. In the longitudinal 10 ft. direction

$$V_{u1} = (L/2 - C/2 - d) \cdot q_u \cdot b$$

$$V_{u1} = (8/2 - 1.167/2 - 19.5/12) \cdot 5.3055 \cdot 6$$

$$V_{u1} = 57.0291 \text{ kips}$$

$$d = V_{u1} / 2\phi \lambda \sqrt{f_c} \cdot b_w$$

$$d = 57.0291 / (2 \times 0.75 \times 1 \times \sqrt{4000} \times 8 \times 6)$$

$$d = 12.524''$$

So,

$$19.5'' > 12.524'' \text{ OKAY}$$

7 Check for Two-Way Shear

Square column

$$b_o = 4(c + d)$$

$$b_o = 4(14 + 19.5)$$

$$b_o = 134 \text{ inch}$$

$$V_{u2} = P_u - q_u(c + d)^2$$

$$V_{u2} = 254.655 - 5.305(1.167 + 19.5/12)^2$$

$$V_{u2} = 254.378 \text{ kips}$$

$$\beta = \text{longer/shorter}$$

$$\beta = 8/6$$

$$\beta = 1.333 < 2$$

$$d_1 = V_{u2} / 4\phi \lambda \sqrt{f_c} \cdot b_o$$

$$d_1 = 254.378 \times 1000 / (4 \times 0.75 \times 1 \times \sqrt{4000} \times 134)$$

$$d_1 = 10.003''$$

$$d_2 = V_{u2} / (\phi(a_s \cdot d_{\text{effective}} / b_o + 2) \lambda_s \lambda \sqrt{f_c} b_o)$$

Where, $\lambda_s = 1$, $a_s = 40$

$$d_2 = 254.378 \times 1000 / (0.75(30 \times 19.5 / 134.016 + 2) \times 1 \times 1 \times \sqrt{4000} \times 134)$$

$$d_2 = 6.286 \text{ in}$$

8 Design Steel Reinforcement In The Longitudinal Direction

$$M_u = q_u/2(L/2 - C/2)^2 \cdot b$$

$$Mu=5.305/2(8/2-1.167/2)^2*6$$

$$Mu=185.785\text{k-ft}$$

$$Ru=Mu/bd^2$$

$$Ru=185.785 \times 1000 \times 12 / 6 \times 12 \times 19.5^2$$

$$Ru=6.785 \text{ psi}$$

$$\rho=0.85 \times 4 / 60 [1 - \sqrt{1 - 2 \times 6.785 / (0.85)(0.9)(4000)}]$$

$$\rho=0.001$$

$$As=\rho b d$$

$$As=0.001 \times 6 \times 12 \times 19.5$$

$$As=1.404 \text{ in}^2$$

$$\text{Min } As_{\text{shrinkage}}=0.0018bh$$

$$\text{Min } As_{\text{shrinkage}}=0.0018 \times 6 \times 12 \times 24$$

$$\text{Min } As_{\text{shrinkage}}=3.1104 \text{ in}^2$$

$$\text{Min } As_{\text{Flexure}}=0.0033bd$$

$$\text{Min } As_{\text{Flexure}}=0.0033 \times 6 \times 12 \times 19.5$$

$$\text{Min } As_{\text{Flexure}}=4.68(\text{SELECTED})$$

For Shorter Side

$$\text{Required Area per foot} = As / \text{Longer side}$$

$$\text{Required Area per foot} = 4.68 / 6$$

$$\text{Required Area per foot} = 0.78 \text{ in}^2 / \text{ft}$$

Spacing

Using # 7 bar

$$\text{Spacing} = Ab / \text{Area per foot} \times 12$$

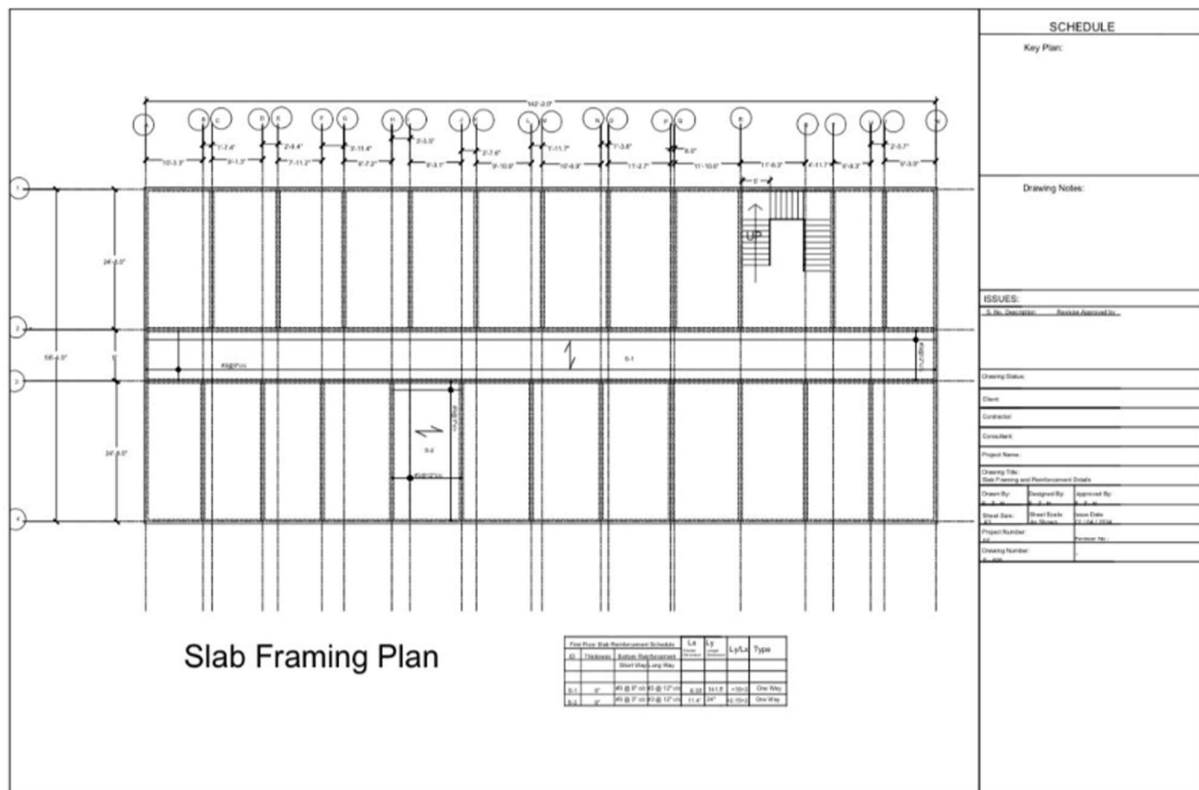
$$\text{Spacing} = 0.6013 / 0.78 \times 12$$

$$\text{Spacing} = 9.251$$

$$\text{Spacing} = 9'' \text{ C/C}$$

SLAB Design:

Slab Framing Plan



METHOD

Known:

- Weight density of concrete = 150 pcf
- Weight density of floor finishes = 144 pcf
- $f_y=60\text{ksi}$
- $f_c=4\text{ksi}$
- Floor finishes = 3"
- Live Load for corridors = 100 psf

Formula:

- Ultimate load = $1.2P_d + 1.6P_L$
- Effective Depth = $h - 1$
- $M_u = w l^2 / 10$
- $R_u = M_u / b d^2$

Design

Simply Supported Slab

- **Factored Loads**

Need to find out first the thickness of the slab

For Simply Supported Slab

$$h(\text{min}) = \text{Span} / 20$$

$$h(\text{min}) = 8.333' / 20$$

$$h(\text{min}) = 5'$$

- **Self-Weight of the Slab**

$$W_s = 5' / 12 \times 150$$

$$W_s = 62.5 \text{ psf}$$

Screeding of 3"

$$W_{sd} = 3' / 12 \times 144$$

$$W_{sd} = 36 \text{ psf}$$

- **Total Dead Loads**

$$\text{Total dead load} = 62.5 + 36$$

$$\text{Total dead load} = 98.5 \text{ psf}$$

$$\text{Live Load for corridor} = 100 \text{ psf}$$

- **Factored Loads**

$$W_u = [1.2(D) + 1.6(L)]$$

$$W_u = [1.2(98.5) + 1.6(100)]$$

$$W_u = 278.2 \text{ psf}$$

$$W_u = 0.2782 \text{ k/ft}$$

- **Maximum Moment**

$$M_u = W_u l^2 / 8$$

$$M_u = (0.2782)(8.333)^2 / 8$$

$$M_u = 2.414 \text{ K-ft/ft}$$

- **Design**

$$M_u = R_u b d^2$$

For R_u

$$R_u = M_u / b d^2$$

Where, $b=12''$ and $d=4''$

$$R_u = 2.414/12 \times (4)^2$$

$$R_u = 0.15 \text{ ksi}$$

$$\rho(\text{req}) = 0.85 f'_c / f_y [1 - \sqrt{1 - 2R_u / \phi * 0.85 * f'_c}]$$

$$\rho(\text{req}) = 0.85 \times 4 / 60 [1 - \sqrt{1 - 2 \times 0.15 / 0.9 (0.85) (4)}]$$

$$\rho(\text{req}) = 0.00284$$

$$\rho(\text{min}) > \rho(\text{req})$$

Taking $\rho(\text{min})$

$$\rho = 200 / f_y$$

$$\rho = (200 / 60000)$$

$$\rho = 0.003$$

$$A_s = \rho b d$$

$$A_s = 0.0033 \times 12 \times 4$$

$$A_s = 0.144 \text{ in}^2/\text{ft}$$

Using # 3 bar

$$\text{Area of \# 3 bar} = 0.11 \text{ in}^2$$

$$S = 12 A_b / A_s$$

$$S = 12 (0.11) / 0.144$$

$$S = 9.16''$$

$$S = 9''$$

Use #3@9''c/c

- **Shrinkage (Secondary) Reinforcement**

$$A_{sh} = 0.0018 b h$$

$$A_{sh} = 0.0018 \times 12 \times 5$$

$$A_{sh} = 0.108 \text{ in}^2$$

Using # 3 bar

$$S = 12 A_b / A_s$$

$$S = 12 (0.11) / 0.108$$

$$S = 12''$$

Use #3@12''c/c

Two end Continuous Slab

$$h(\text{min}) = \text{Span} / 28$$

$$h(\text{min}) = 11.88'' / 20$$

$$h(\text{min}) = 4.96'' = 5''$$

- **Moment from ETABS**

$$M_u = 5.619 \text{ k-ft/ft}$$

- **Design**

$$\mu = R_u b d^2$$

For R_u

$$R_u = \mu / b d^2$$

Where, $b = 12''$ (Unit Width) and $d = 5 - 1 = 4''$

$$R_u = 5.619 / 12 \times (4)^2 \times 12$$

$$R_u = 0.351 \text{ ksi}$$

$$\rho (\text{req}) = 0.85 f'_c / f_y [1 - \sqrt{1 - 2 R_u / \phi * 0.85 * f'_c}]$$

$$\rho (\text{req}) = 0.85 \times 4 / 60 [1 - \sqrt{1 - 2 \times 0.351 / 0.9 (0.85) (4)}]$$

$$\rho (\text{req}) = 0.007$$

$$\rho = 200 / f_y$$

$$\rho = (200 / 60000)$$

$$\rho = 0.003$$

$$\rho (\text{min}) < \rho (\text{req})$$

That's why Taking $\rho (\text{req})$

$$A_s = \rho b d$$

$$A_s = 0.007 \times 12 \times 4$$

$$A_s = 0.336 \text{ in}^2$$

Using # 3 bar

$$\text{Area of \# 3 bar} = 0.11 \text{ in}^2$$

$$S = 12 A_b / A_s$$

$$S = 12 (0.11) / 0.336$$

$$S = 3.053''$$

$$S = 3''$$

Use #3@3''c/c

- **Shrinkage (Secondary) Reinforcement**

$$A_{sh} = 0.0018 b h$$

$$A_{sh} = 0.0018 \times 12 \times 5$$

$$A_{sh} = 0.108 \text{ in}^2$$

Using # 3 bar

$$S = 12 A_b / A_s$$

$$S = 12 (0.11) / 0.108$$

$$S = 12''$$

Use #3@12''c/c

Solution By Excel

Corner, Interior and Edge Footing Design

DESIGN OF INTERIOR FOOTING

DATA	UNITS	VALUES	FORMULAS
Thickness of Columns	inches	14	(USE CELL D4B)
Width of Columns	inches	14	(USE CELL D4B)
Height of Columns	ft	10.0	(USE CELL D5D)
at (for columns)	ft	10.0	
Assume Longer side of footing (for 1st way)	ft	7	
Assume shorter side of footing (for 2nd way)	ft	7	
Concrete strength (psi)	psi	4000	
Steel strength (ksi)	ksi	60	
Concrete weight (pcf)	pcf	150	
Steel weight (pcf)	pcf	490	
Dead Load (kips/ft)	k/ft	10.0	
Live Load (kips/ft)	k/ft	10.0	
Total Load (kips/ft)	k/ft	20.0	
Assume total depth of footing (ft)	ft	3	

Effective Soil Pressure

DATA	UNITS	VALUES	FORMULAS
1st way of footing	psi	10.0	
2nd way of footing	psi	10.0	
Effective soil pressure	psi	10.0	

Area of Footing

DATA	UNITS	VALUES	FORMULAS
Total required area	sq ft	10.0	
Area of footing	sq ft	10.0	
Effective width	ft	10.0	
Effective length	ft	10.0	

Check for One-Way Shear

DATA	UNITS	VALUES	FORMULAS
Clear cover of footing	inches	4	
Effective depth	inches	24	
Effective length	ft	10.0	
Effective width	ft	10.0	
Effective area	sq ft	10.0	
Effective length	ft	10.0	
Effective width	ft	10.0	
Effective area	sq ft	10.0	

DESIGN OF CORNER & EDGE FOOTING

DATA	UNITS	VALUES	FORMULAS
Thickness of Columns	inches	14	(USE CELL D4B)
Width of Columns	inches	14	(USE CELL D4B)
Height of Columns	ft	10.0	(USE CELL D5D)
at (for columns)	ft	10.0	
Assume Longer side of footing (for 1st way)	ft	7	
Assume shorter side of footing (for 2nd way)	ft	7	
Concrete strength (psi)	psi	4000	
Steel strength (ksi)	ksi	60	
Concrete weight (pcf)	pcf	150	
Steel weight (pcf)	pcf	490	
Dead Load (kips/ft)	k/ft	10.0	
Live Load (kips/ft)	k/ft	10.0	
Total Load (kips/ft)	k/ft	20.0	
Assume total depth of footing (ft)	ft	3	

Effective Soil Pressure

DATA	UNITS	VALUES	FORMULAS
1st way of footing	psi	10.0	
2nd way of footing	psi	10.0	
Effective soil pressure	psi	10.0	

Area of Footing

DATA	UNITS	VALUES	FORMULAS
Total required area	sq ft	10.0	
Area of footing	sq ft	10.0	
Effective width	ft	10.0	
Effective length	ft	10.0	

Check for One-Way Shear

DATA	UNITS	VALUES	FORMULAS
Clear cover of footing	inches	4	
Effective depth	inches	24	
Effective length	ft	10.0	
Effective width	ft	10.0	
Effective area	sq ft	10.0	
Effective length	ft	10.0	
Effective width	ft	10.0	
Effective area	sq ft	10.0	

Slab Design

DESIGN OF SLAB

Simply Supported Slab

DATA	UNITS	VALUES	FORMULAS
Specific weight of RCC	pcf	150	
Specific weight of PCC	pcf	140	
Finishes	inches	3	
Length of Slab	ft	141.5	
Width of Slab	ft	8.33	
Thickness of slab	inches	5	
Self weight of slab	pcf	62.5	
Reduction factor		0.85	
Strength of steel (fy)	ksi	60	
Strength of concrete (fc)	ksi	4	
Dead Load of Finishes	pcf	36	
Self Weight of Slab	pcf	62.5	
Finishes	inches	5	
Finishes Load	pcf	100	
Total Dead Load	pcf	98.5	
Ultimate Load	k/ft	0.2782	
Maximum Moment (Mu)	ft-k	2.41299898	
Mu = Ru bd^2			
b (inches)	inches	12	
d (inches)	inches	0.151	
Moment of resistance (Ru)	ksi	0.00260260	
FOR MAIN BARS			
Area (in^2)	in^2	0.003	
As (in^2)	in^2	0.100	
As (in^2)	in^2	8.000	
FOR DISTRIBUTION BARS			
As (in^2)	in^2	0.1080	
As (in^2)	in^2	12.000	
Main Bars Reinforcement	#3@12" c/c		
Distribution Bars Reinforcement	#3@12" c/c		

Two end continuous

Minimum Thickness Required by ACI for Different Types of Slab

DATA	UNITS	VALUES	FORMULAS
Minimum Thickness	inches	5.5	
Two end continuous	Span/28		
Thickness of slab is	inches	5	
Length of Slab	ft	141.5	
Width of Slab	ft	8.33	
Mu = Ru bd^2			
b (inches)	inches	12	
d (inches)	inches	0.151	
Moment of resistance (Ru)	ksi	0.00260260	
FOR MAIN BARS			
Area (in^2)	in^2	0.003	
As (in^2)	in^2	0.100	
As (in^2)	in^2	8.000	
FOR DISTRIBUTION BARS			
As (in^2)	in^2	0.1080	
As (in^2)	in^2	12.000	
Main Bars Reinforcement	#3@12" c/c		
Distribution Bars Reinforcement	#3@12" c/c		

Final ETABS Output and Structural Design Validation

After completing the ETABS analysis, all structural members, including slabs, columns, and beams, passed successfully without any severe failures. This confirms that the design for our G+3 structure is robust and suitable for real-time implementation.

Analysis Overview:

1. ETABS Analysis Results:

The analysis was conducted under ultimate load conditions as per the ACI-318-09 Code. All members demonstrated sufficient strength and stability, with no significant design violations or overstressed elements.

The output drawings obtained from ETABS, showcasing reinforcement details for columns and beams, aligned perfectly with the design requirements and reflected the original design intent.

2. Load Testing:

The structure was subjected to ultimate load testing in ETABS, simulating realistic conditions to ensure compliance with design standards.

Design Validation Through Multiple Approaches:

To ensure the accuracy and reliability of the design, several methods were employed:

1. Manual Calculations:

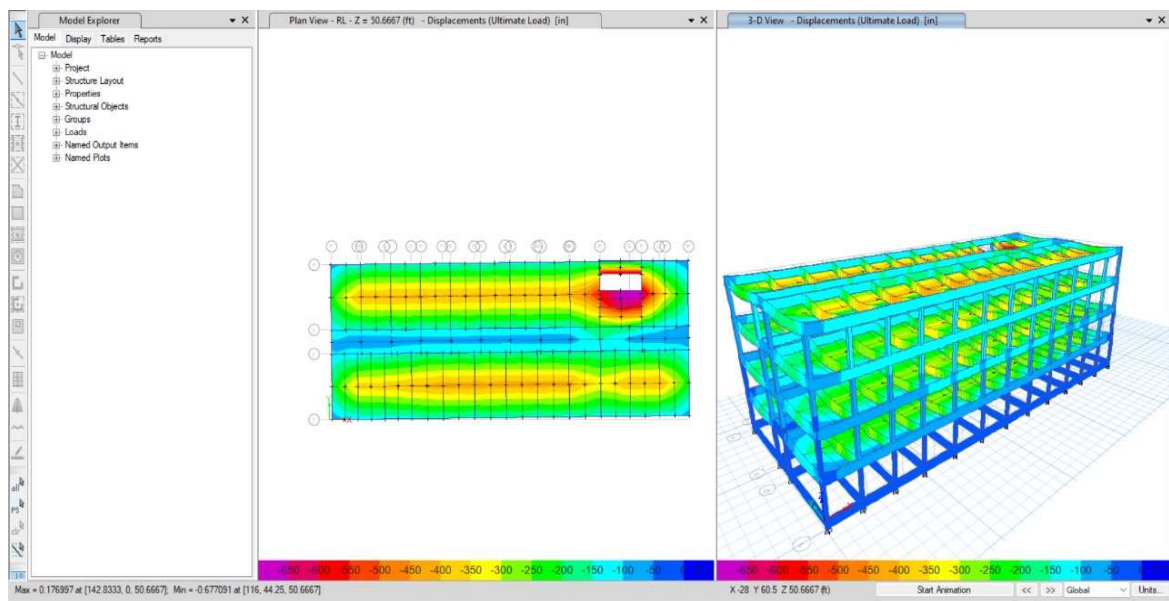
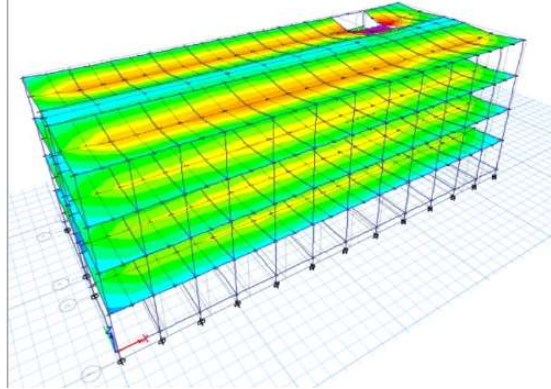
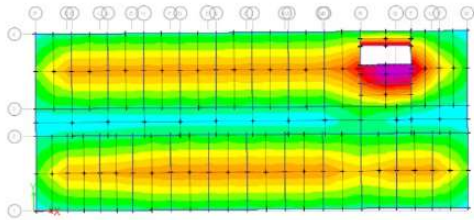
- Detailed calculations were performed manually to cross-verify the design parameters.
- These computations covered critical structural elements, including slabs, columns, and beams, and adhered strictly to the governing codes.

2. Excel-Based Calculations:

- Spreadsheet tools were used to simplify and validate repetitive calculations.
- This approach ensured consistency and helped confirm the values obtained manually and through ETABS.

3. ETABS Software Validation:

- Advanced finite element analysis was conducted using ETABS to simulate the structural performance under various load combinations.
- The analysis reinforced the validity of the manual and Excel-based results, with minor, negligible differences attributed to rounding or modeling assumptions.



Observations and Findings:

- Across all approaches, the ultimate loads and design outputs were nearly identical.
- The differences in values were minimal, showcasing the precision and reliability of our design methodology.
- This consistency among manual, Excel-based, and software-derived results further demonstrates that the structure is well-optimized and ready for implementation.

Conclusion:

The G+3 structure was rigorously tested and validated using a combination of manual calculations, spreadsheet analysis, and ETABS software simulations. The uniformity in the results across all methods confirms the structural integrity and suitability of the design for real-time applications.

All necessary code provisions and design standards were adhered to, and the analysis outputs reflect a strong, stable, and efficient structure. Therefore, this design can be confidently implemented without hesitation.

SCHEDULE

Key Plan:

DRAWING NOTES

1. DOORS 4'X7'
2. WINDOWS 6'x4'
3. ventilator 17'x2'
4. Riser 6"
5. Tread 1"
6. Wall thickness 6"

Construction Drawings

Project Location:
N+1 Block Civil Eng Dept. NEDUET

Drawing Title:
Submission Drawing N-Block G+1

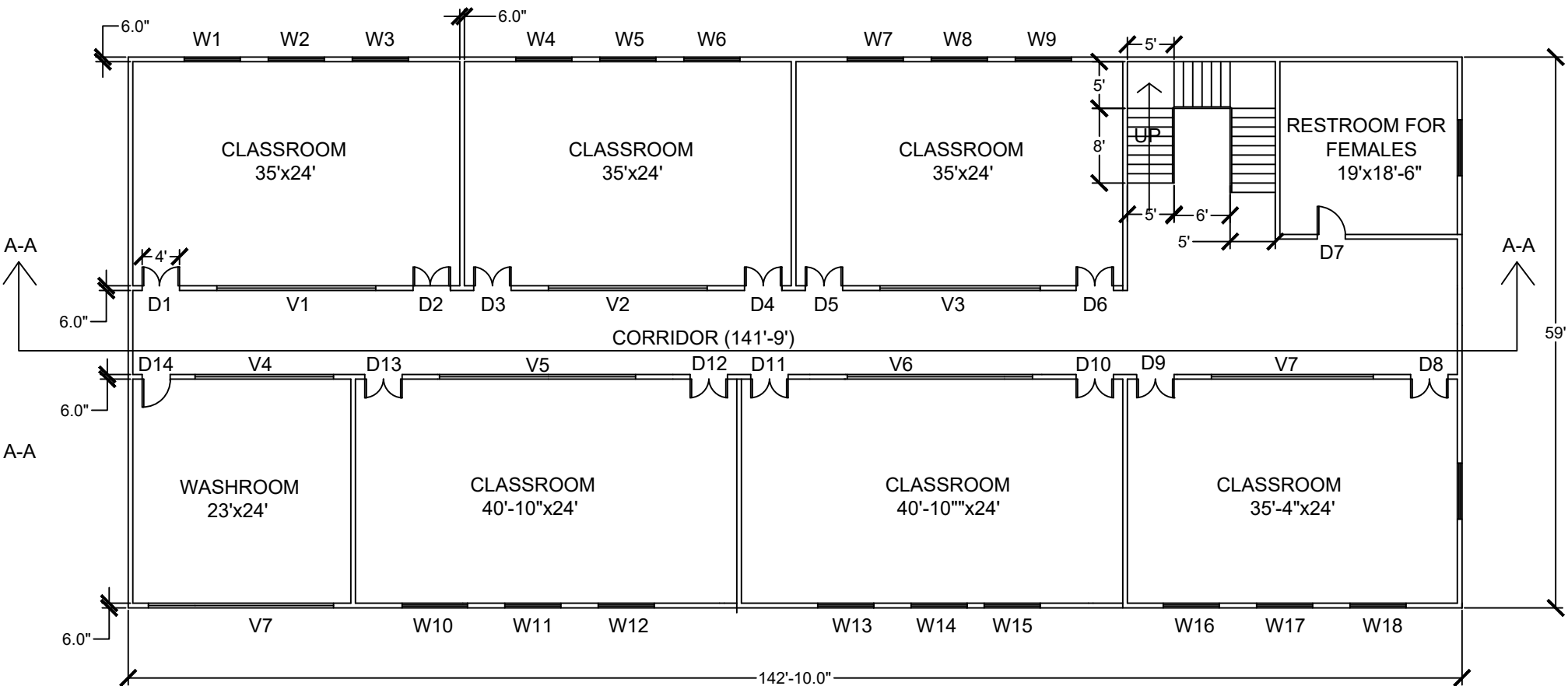
- Group Members:
1. Muhammad Aqib Hameed
(CE-22011)
 2. Hafiz Muhammad Yaseen
(CE-22014)
 3. Muhammad Zaryab Ali
(CE22016)

Scale:

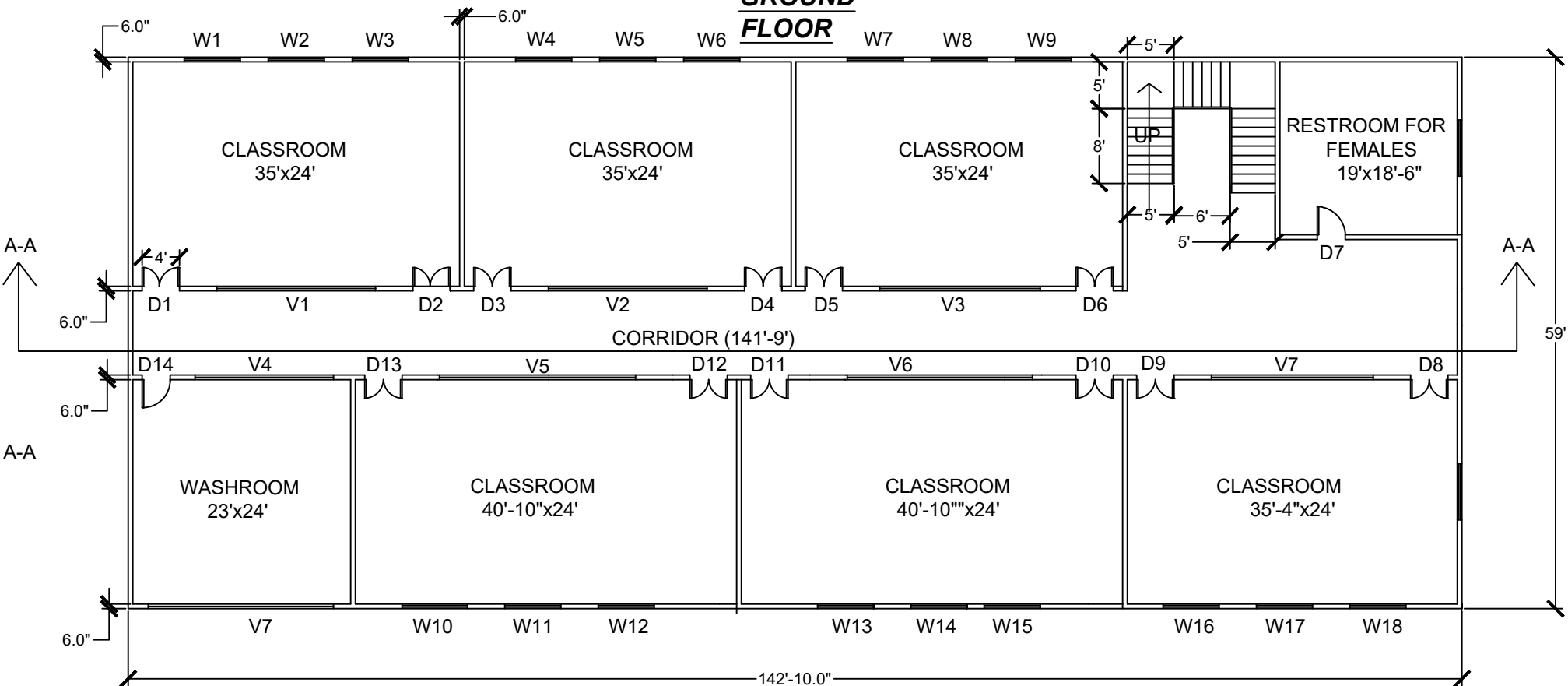
$\frac{1}{16}"=1'-0"$

Sheet Size:

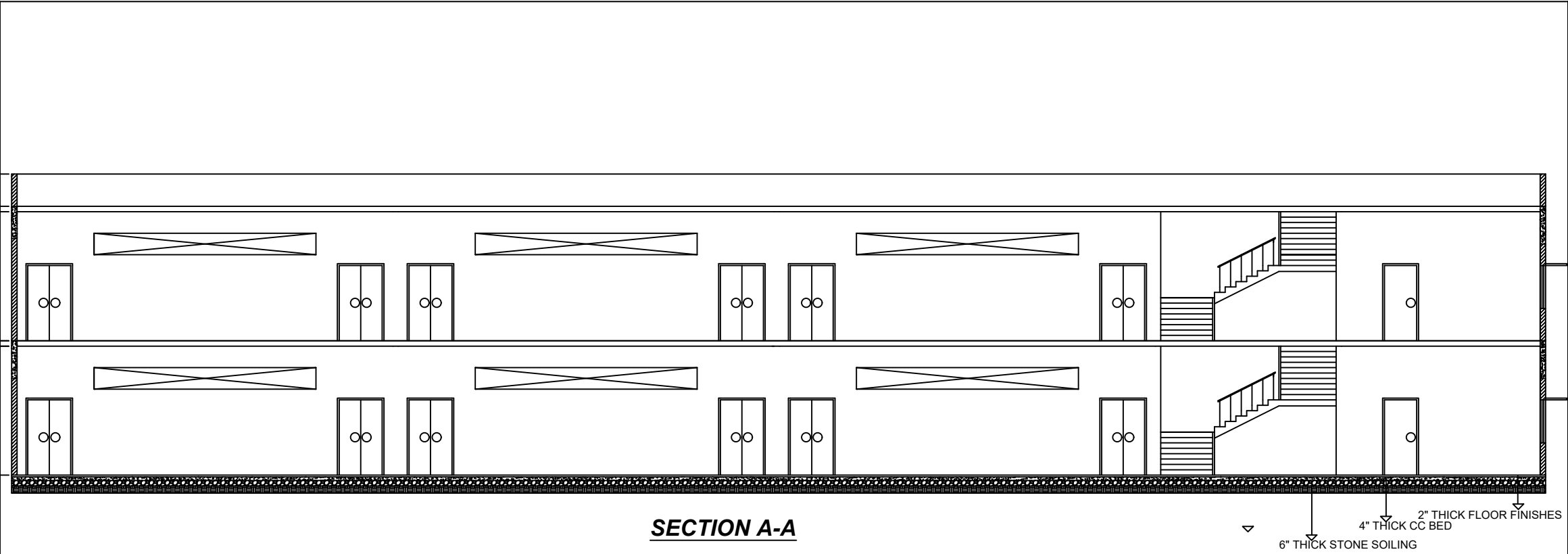
A-3



GROUND FLOOR



FIRST FLOOR

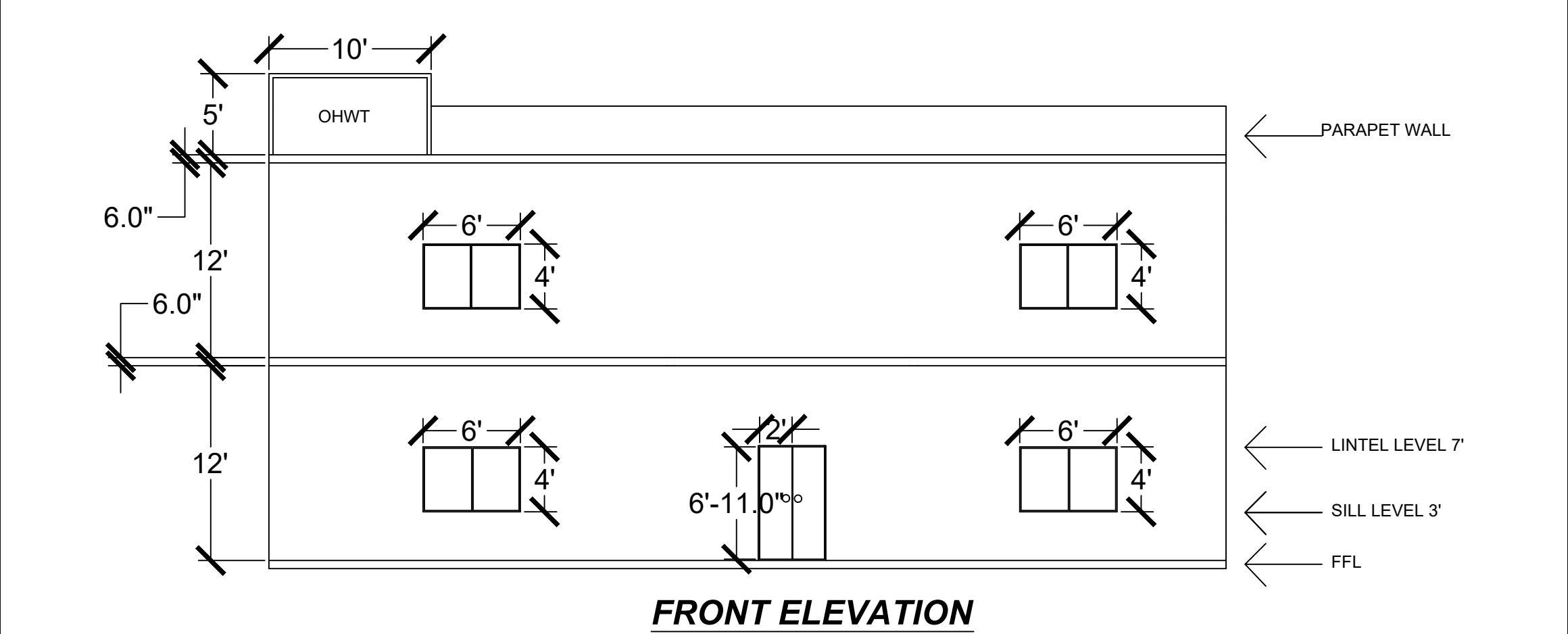


SCHEDULE

Key Plan:

DRAWING NOTES

- 1. DOORS 4'X7'
- 2. WINDOWS 6'x4'
- 3. ventilator 17'x2'
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Construction Drawings

Project Location:
N+1 Block Civil Eng Dept. NEDUET

Drawing Title:
Submission Drawing N-Block G+1

- Group Members:
- 1. Muhammad Aqib Hameed (CE-22011)
 - 2. Hafiz Muhammad Yaseen (CE-22014)
 - 3. Muhammad Zaryab Ali (CE22016)

Scale:

1/16"=1'-0"

Sheet Size:

A-3

Figure 10 is an elevation view of the bridge deck, showing the deck, parapets, and reinforcement details. The drawing includes dimensions for the deck width (296 in), parapet width (78.5 in), and reinforcement spacing (#3 @ 6 in). The reinforcement is shown as #3 bars at 6 in spacing, with a total length of 165.5 in for the deck and 166.5 in for the parapets. The deck is shown with a width of 296 in, and the parapets are shown with a width of 78.5 in. The reinforcement is shown as #3 bars at 6 in spacing, with a total length of 165.5 in for the deck and 166.5 in for the parapets. The deck is shown with a width of 296 in, and the parapets are shown with a width of 78.5 in. The reinforcement is shown as #3 bars at 6 in spacing, with a total length of 165.5 in for the deck and 166.5 in for the parapets.

[illegible]

1

77in 166in 166in 77in 108in 77in 166in 166in 77in

#3@6 #3@16 #3@6 #3@6 #3@6 #3@16 #3@6

296 in 108 in 296 in

[illegible]

Technical drawing of a mechanical part, likely a bracket or support, showing dimensions and features. The part is L-shaped with a vertical section and a horizontal section. The vertical section has a width of 0.32 in. and a height of 0.88 in. The horizontal section has a width of 0.88 in. and a height of 0.32 in. The part has four mounting holes (two on the vertical section, two on the horizontal section) and a central slot. The drawing is labeled with dimensions and part numbers.

CONTINUOUS BEAM SECTION

Key Plan:

Drawing Notes:

ISSUES:

S. No.	Description	Revision	Approved by
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Drawing Status:

Client:

Contractor:

Consultant:

Project Name:

Drawing Title:
Slab Framing and Reinforcement Details

Drawn By: F Z H	Designed By: F Z H	Approved By: F Z H
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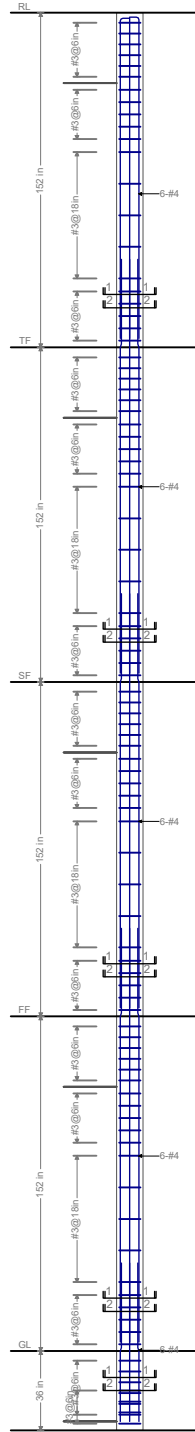
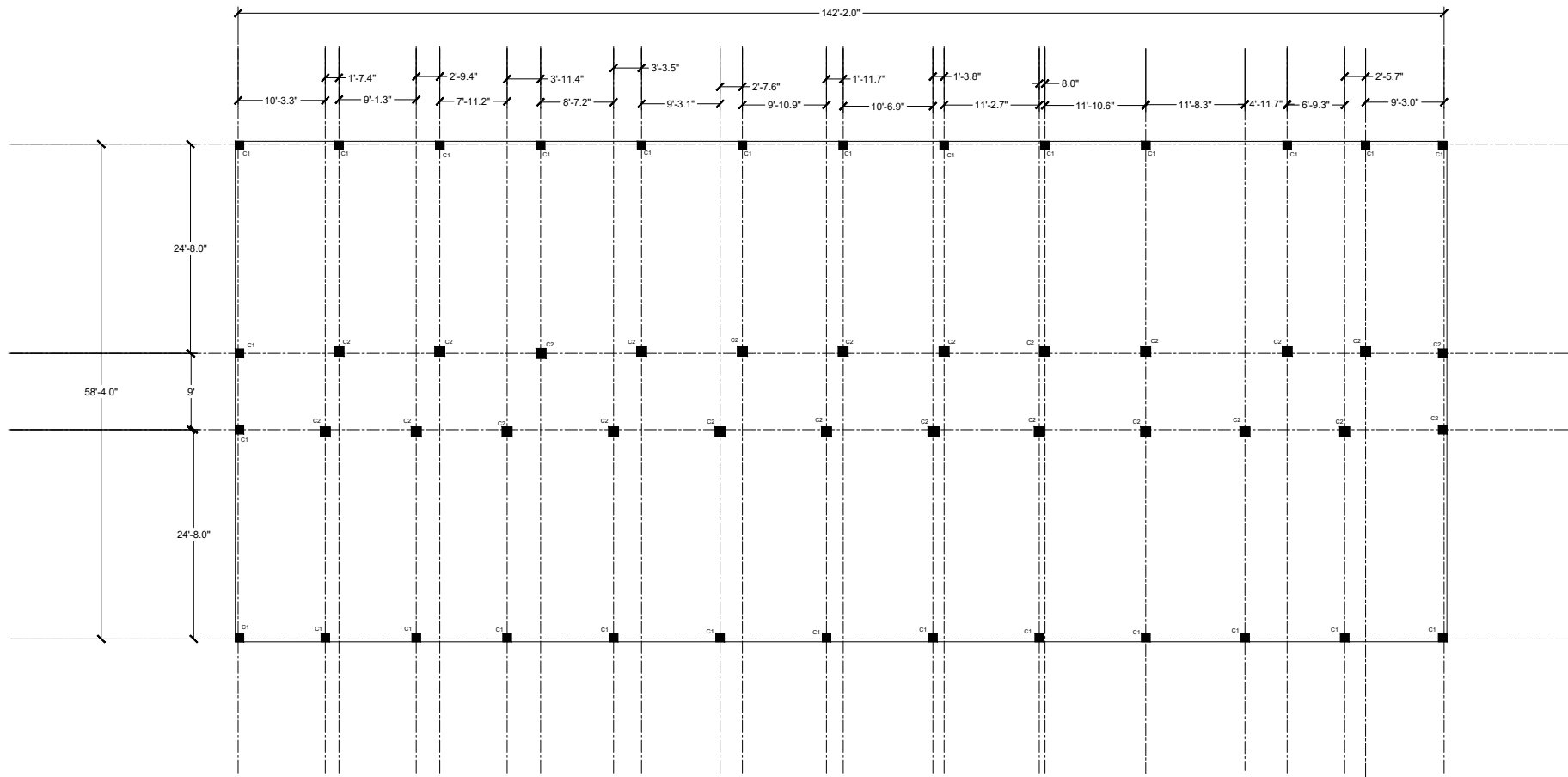
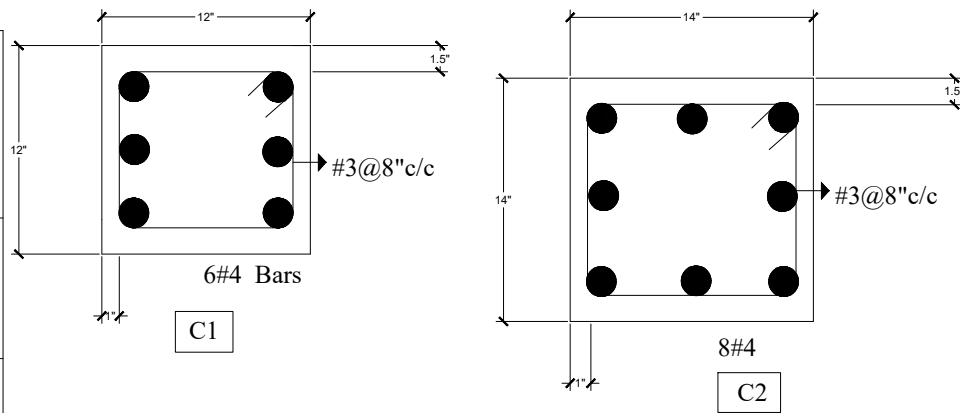
Sheet Size: A3	Sheet Scale: As Shown	Issue Date 02 / 04 / 2024
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Project Number:	Revision No.:
07	

Drawing Number:
S - 605

Reinforcment Schdule For Column

Column I.D	Column Cross Section	Numbers	Main Reinforcement Bars	Tie Bar
C1	12'X12'	30	6#4	#3@8"c/c
C2	14'X14'	22	10#4	#3@8"c/c



CS-01:Full Elevation-2
(Scale = 1:4)

COLUMN Layout Plan

SCHEDULE

Key Plan:

Drawing Notes:

ISSUES:

S. No. Description Revision Approved by

Drawing Status:

Client:

Contractor:

Consultant:

Project Name:

Drawing Title:
Slab Framing and Reinforcement Details

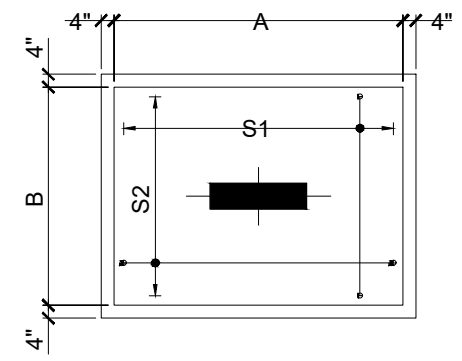
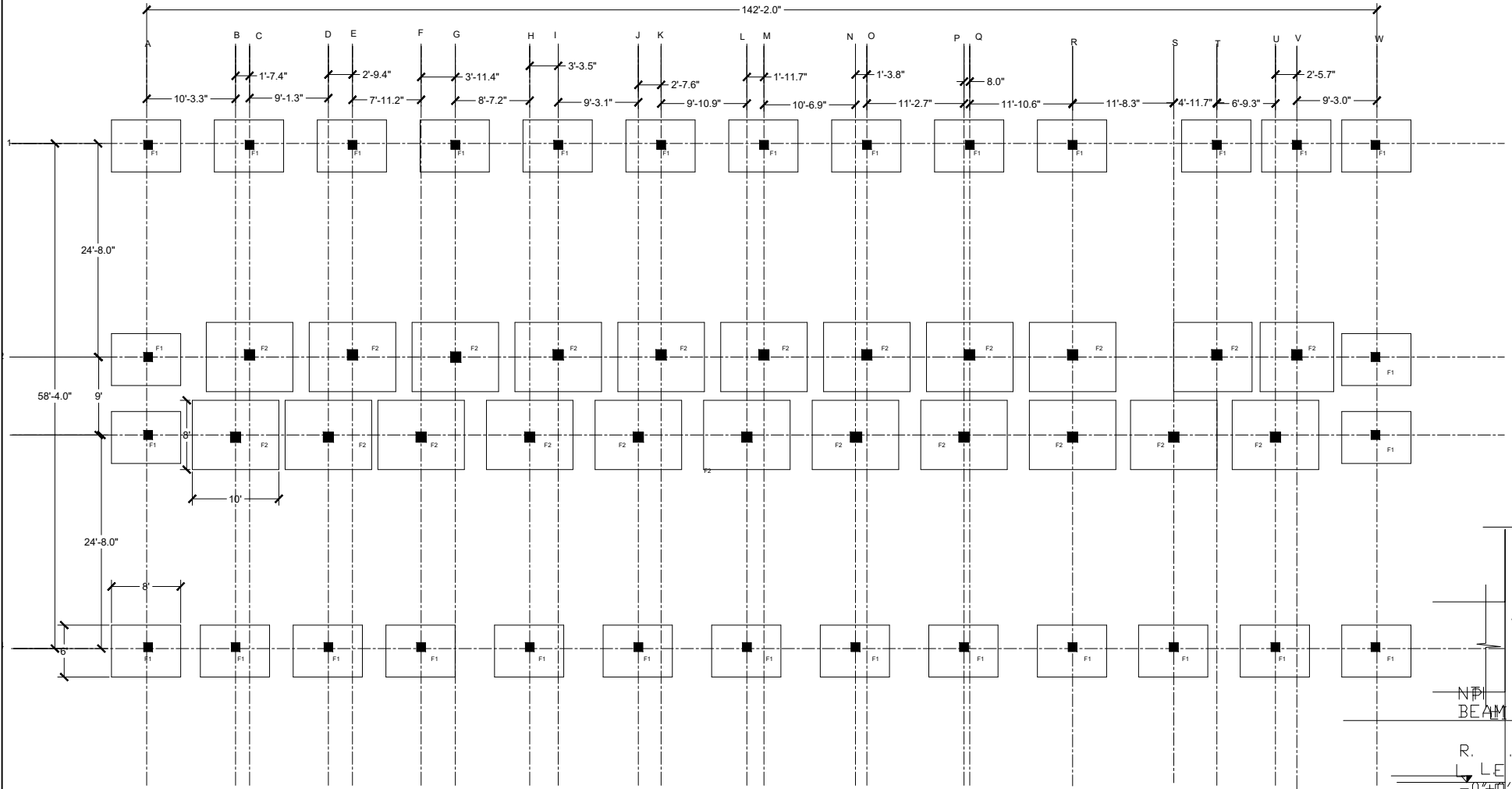
Drawn By: F Z H Designed By: F Z H Approved By: F Z H

Sheet Size: A3 Sheet Scale: As Shown Issue Date: 02 / 04 / 2024

Project Number: 07 Revision No.:

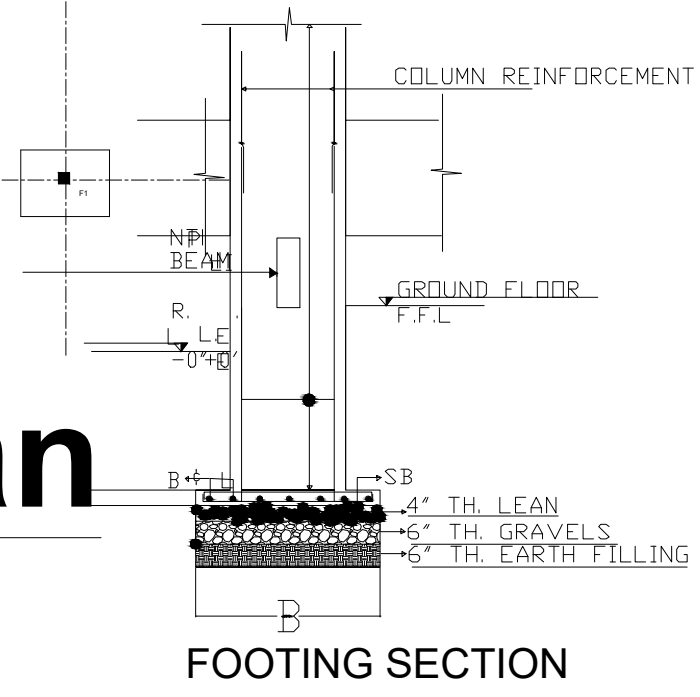
Drawing Number: S - 605

Foundation Reinforcement Details					
ID	Size	Bottom Reinforcement		Top Reinforcement	
		S1	S2	S1'	S2'
F-1	6'x8'		#7@ 9" c/c	-	-
F-2	8'x10'	#7@ 9" c/c	#7@ 9" c/c	#7@ 9" c/c	#7@ 9" c/c



ISOLATED FOOTING

Foundation Layout Plan



SCHEDULE

Key Plan:

Drawing Notes:

ISSUES:

S. No.	Description	Revision	Approved by
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Drawing Status:

Client:

Contractor:

Consultant:

Project Name:

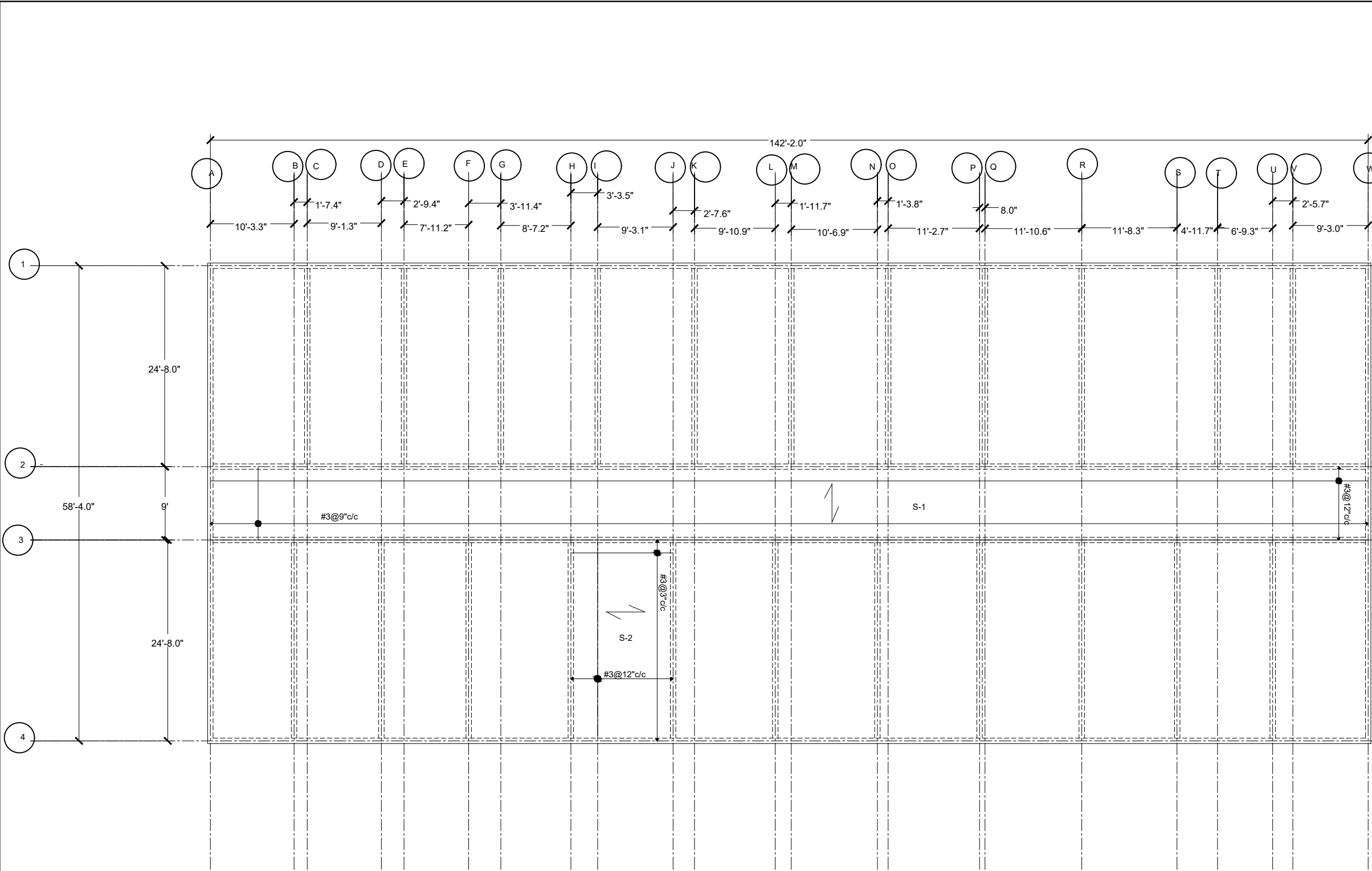
Drawing Title:
Slab Framing and Reinforcement Details

Drawn By: F Z H	Designed By: F Z H	Approved By: F Z H
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Sheet Size: A3	Sheet Scale: As Shown	Issue Date: 02 / 04 / 2024
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Project Number: 07	Revision No.: -
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Drawing Number: S - 605	
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Slab Framing Plan

First Floor Slab Reinforcement Schedule				Lx	Ly	Ly/Lx	Type
ID	Thickness	Bottom Reinforcement		Shorter Dimension	Longer Dimension		
		Short Way	Long Way				
S-1	6"	#3 @ 9" c/c	#3 @ 12" c/c	8.33'	141.5'	=16>2	One Way
S-2	6"	#3 @ 3" c/c	#3 @ 12" c/c	11.4'	24"	=2.15>2	One Way

SCHEDULE

Key Plan:

Drawing Notes:

ISSUES:

S. No.	Description	Revision	Approved by
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Drawing Status:

Client:

Contractor:

Consultant:

Project Name:

Drawing Title:
Slab Framing and Reinforcement Details

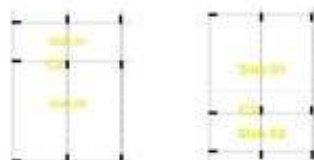
Drawn By: F Z H	Designed By: F Z H	Approved By: F Z H
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Sheet Size: A3	Sheet Scale: As Shown	Issue Date: 02 / 04 / 2024
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Project Number: 07	Revision No.: -
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Drawing Number: S - 605	
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Submitted: 10/10/2013; Accepted: 01/09/2014; Published: 02/09/2014

[illegible][illegible]

DOWN OF CORNER & EDGE COLLAPSE

Equipment			
Description	Quantity	Unit Price	Total Cost
Excavator	1	10000	10000
Truck	1	5000	5000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1	8000	8000
Grader	1	6000	6000
Compactor	1	3000	3000
Drill	1	4000	4000
Crane	1	12000	12000
Concrete Mixer	1	7000	7000
Batching Plant	1	15000	15000
Storage Silo	1	10000	10000
Water Tank	1	8000	8000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1	8000	8000
Grader	1	6000	6000
Compactor	1	3000	3000
Drill	1	4000	4000
Crane	1	12000	12000
Concrete Mixer	1	7000	7000
Batching Plant	1	15000	15000
Storage Silo	1	10000	10000
Water Tank	1	8000	8000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1	8000	8000
Grader	1	6000	6000
Compactor	1	3000	3000
Drill	1	4000	4000
Crane	1	12000	12000
Concrete Mixer	1	7000	7000
Batching Plant	1	15000	15000
Storage Silo	1	10000	10000
Water Tank	1	8000	8000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1	8000	8000
Grader	1	6000	6000
Compactor	1	3000	3000
Drill	1	4000	4000
Crane	1	12000	12000
Concrete Mixer	1	7000	7000
Batching Plant	1	15000	15000
Storage Silo	1	10000	10000
Water Tank	1	8000	8000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1	8000	8000
Grader	1	6000	6000
Compactor	1	3000	3000
Drill	1	4000	4000
Crane	1	12000	12000
Concrete Mixer	1	7000	7000
Batching Plant	1	15000	15000
Storage Silo	1	10000	10000
Water Tank	1	8000	8000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1	8000	8000
Grader	1	6000	6000
Compactor	1	3000	3000
Drill	1	4000	4000
Crane	1	12000	12000
Concrete Mixer	1	7000	7000
Batching Plant	1	15000	15000
Storage Silo	1	10000	10000
Water Tank	1	8000	8000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1	8000	8000
Grader	1	6000	6000
Compactor	1	3000	3000
Drill	1	4000	4000
Crane	1	12000	12000
Concrete Mixer	1	7000	7000
Batching Plant	1	15000	15000
Storage Silo	1	10000	10000
Water Tank	1	8000	8000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1	8000	8000
Grader	1	6000	6000
Compactor	1	3000	3000
Drill	1	4000	4000
Crane	1	12000	12000
Concrete Mixer	1	7000	7000
Batching Plant	1	15000	15000
Storage Silo	1	10000	10000
Water Tank	1	8000	8000
Generator	1	2000	2000
Water Pump	1	1500	1500
Backhoe Loader	1		

EXPOSURE OF HYDROLYZABLE POLYURETHANES

Name	ID	Score
John Doe	101	85
Jane Smith	102	78
Michael Brown	103	92
Sarah Jones	104	88
David Wilson	105	75
Emily Davis	106	82
James Miller	107	79
Olivia Taylor	108	90
Benjamin Clark	109	87
Mia White	110	76
Lucas Green	111	83
Ava Black	112	77
Isabella Gray	113	89
Robert King	114	74
Charlotte Lee	115	86
William Hall	116	73
Amelia Scott	117	91
Thomas Young	118	84
Harper Adams	119	72
Sebastian Baker	120	81

CHANGING THE PASTORAL ROLE

Financial Summary			
Category	Amount	Unit	Notes
Revenue	1000000	USD	
Expenses	800000	USD	
Profit	200000	USD	
Assets	500000	USD	
Liabilities	300000	USD	
Equity	200000	USD	
Financial Ratios			
Current Ratio	1.67		
Debt to Equity Ratio	1.5		
Return on Assets	40%		
Return on Equity	100%		
Financial Projections			
Year	2020	2021	2022
Revenue	1000000	1100000	1200000
Expenses	800000	850000	900000
Profit	200000	250000	300000
Assets	500000	550000	600000
Liabilities	300000	320000	340000
Equity	200000	230000	260000
Financial Ratios (2022)			
Current Ratio	1.76		
Debt to Equity Ratio	1.29		
Return on Assets	50%		
Return on Equity	115%		
Financial Projections (2023)			
Year	2023	2024	2025
Revenue	1300000	1400000	1500000
Expenses	950000	1000000	1050000
Profit	350000	400000	450000
Assets	650000	700000	750000
Liabilities	400000	420000	440000
Equity	250000	280000	310000
Financial Ratios (2025)			
Current Ratio	1.93		
Debt to Equity Ratio	1.03		
Return on Assets	67%		
Return on Equity	145%		

EXPOSURE TO CRYSTALLINE SILICA DUST

[illegible]

CONCERN OF THE

Project Schedule

1999 2000

Project Kick-off
Project Planning
Project Execution
Project Closeout

Project Schedule (continued)

1999 2000

Project Kick-off
Project Planning
Project Execution
Project Closeout

CHROMATOPHORE BEHAVIOR ON HPLC SOLVENTS

[illegible]

[CROCHET TOP SLIM](#)

The screenshot displays a complex spreadsheet with multiple tables and data entries. The tables are organized into sections, with some having headers like 'Amount' and 'Date'. The data includes various numerical values and dates, such as '1/1/2010', '1/1/2011', and '1/1/2012'. The tables are color-coded, with some having yellow headers and others having blue headers. The overall layout is dense and appears to be a detailed financial or operational report.